

SUB-RECIPIENT AGREEMENT

This SUB-RECIPIENT AGREEMENT (this “Agreement”) is made as of September 24, 2024 by and between National Center for Defense Manufacturing and Machining having offices at 699 Scalp Avenue, Johnstown, PA 15904-1619 (“NCDMM”) and Youngstown Edison Business Incubator, having offices at 241 West Federal Street, Youngstown, OH 44503 (“Sub-Recipient”) with a Unique Entity ID (UEI) number of E38PN6F4AVU3 and a Cage Code of 5EAR9.

WITNESSETH:

WHEREAS, NCDMM is a manufacturing technology center that has entered an agreement with the United States Department of the Air Force, dated as of November 25, 2019 under cooperative agreement number FA8650-20-2-5700 with CFDA number 12.800, to act as an administrator and technical performer in the fulfilment of the “America Makes Partnership (AMP) The National Additive Manufacturing Innovation Institute” (the “Prime Agreement”);

WHEREAS, NCDMM desires to engage Sub-Recipient and America Makes (NAMII) member to assist NCDMM in satisfying certain NCDMM obligations under the Prime Agreement; and

WHEREAS, the activities contemplated by this Agreement are of mutual interest and benefit to NCDMM and Sub-Recipient, and will further the fulfilment of America Makes (NAMII);

NOW, THEREFORE, in consideration of the mutual covenants and agreements contained herein, NCDMM and Sub-Recipient hereby agree as follows:

ARTICLE 1. DEFINITIONS AND SCHEDULES

1.1 **Definitions.** In addition to the terms defined elsewhere in this Agreement, the following terms shall have the meanings set forth below:

1.1.1 **“Affiliate”** means, with respect to any person or entity, any other person or entity directly or indirectly controlling, controlled by, or under common control with, such first person or entity. For purposes of this definition, the term “control” (including the correlative meanings of the terms “controlled by” and “under common control with”), as used with respect to any person or entity, shall mean the possession, directly or indirectly, of the power to direct or cause the direction of the management or policies of such person or entity, whether through the ownership of voting securities or by agreement or otherwise.

1.1.2 **“Applicable Law”** means all applicable federal, state and local laws, rules, regulations and guidelines relating to the conduct of the Parties’ businesses and the performance by the Parties of their respective obligations under this Agreement.

1.1.3 **“Exploit”** means to make, have made, import, use, sell, offer for sale or otherwise dispose of, including any research, development, registration, modification, enhancement, improvement, manufacture, storage, formulation, optimization, export, transport, distribution, promotion or marketing related thereto.

1.1.5 **“Facility”** means the facility of Sub-Recipient, or any other facility used for the conduct of the activities that has been approved by NCDMM and complies with Applicable Law

1.1.6 **“Invention”** means any discovery, improvement, process, formula, invention, know-how, trade secret, procedure, device, or other intellectual property, whether or not patentable.

1.1.7 **“Materials”** means the equipment and materials supplied to Sub-Recipient by or on behalf of, or purchased at the expense of, NCDMM, reasonably necessary for Sub-Recipient to perform activities, solely as set forth in the SOW.

1.1.8 **“Party”** means either NCDMM or Sub-Recipient; **“Parties”** means both NCDMM and Sub-Recipient.

1.1.9 **“Statement of Work”** means the plan set out in Schedule A hereto.

1.1.10 **“Consortium Research”** means research supported by membership dues and / or Incorporates funds provided by the FA8650-20-2-5700 cooperative agreement. Consortium Research does not include:

1.1.10.1 Research carried out under separate contract unless specified therein, or

1.1.10.2 Research fully funded by a Member and/or without using any of the funds specified in 1.1.10 above, or

1.1.10.3 Research carried out independently by a Member using America Makes (NAMII) facilities

1.1.11 **“Consortium Developed Intellectual Property”** means individually and collectively all Intellectual Property which are conceived or made solely or jointly by Member(s) during the performance of this sub award project under Consortium Research.

1.1.12 **“Background Intellectual Property”** means technical know-how, inventions, technical data, discoveries, materials, samples, software, software programs, whether patentable or not, copyrightable programs, documentation and reports whether in existence at the time of this agreement or coming into existence subsequent to this agreement, which were not developed in the course of performance of any sub award under the America Makes (NAMII) consortium.

1.1.13 **“Intellectual Property”** means technical know-how, inventions, technical data, discoveries, materials samples, software, software programs, documentation, reports, any and all other copyrightable materials, and/or invention or discovery that is or may be patentable or otherwise protectable under title 35 of the U.S. Code.

1.2 Schedules. The Schedules attached hereto are incorporated in and are deemed to be an integral part of this Agreement.

ARTICLE 2. ACTIVITIES

2.1 . Sub-Recipient shall undertake and perform work in accordance with the Statement of Work (SOW) attached hereto as Schedule A, as the same may be amended from time to time in accordance with the terms hereof (Sub-Recipient’s responsibilities as set out in the SOW being hereinafter referred to as the “Activities”). The specific objectives, scope, deliverables and estimated schedule for the Activities are set forth in the SOW. Sub-Recipient agrees that it shall implement changes to the SOW that are reasonably requested in writing by NCDMM, subject, if applicable, to NCDMM’s payment of any reasonable additional fees and/or expenses incurred by Sub-Recipient as determined in accordance with Section 4.4 that arise solely from such changes to the SOW. Any material change in the SOW shall be evidenced in a written amendment of Schedule A.

2.2 Cooperation. The Parties shall cooperate with each other in the performance of this Agreement and shall deal honestly and in good faith with each other.

2.3 Reporting/Transfer of Results. Sub-Recipient shall comply with the monthly written progress report requirements set forth in Section 4.5.1 and, if requested by NCDMM, on an ongoing basis, Sub-Recipient shall keep NCDMM informed of progress and interim results on an informal basis, including if requested, periodic meetings to discuss Activities progress and interim results. Further specific guidance regarding programmatic requirements are found in Schedule D. Upon completion of the Activities, termination of this Agreement pursuant to Section 10.2, or expiration of this Agreement (without a further agreement related to the substance hereof entered into as contemplated in Section 10.1), Sub-Recipient will provide NCDMM with a comprehensive written report of the results that have been developed, compiled or learned during the course of the Activities, and will comply with NCDMM's reasonable requests for follow-up information.

ARTICLE 3. SUPPLY OF MATERIALS AND INFORMATION

3.1 Supply or Purchase of Materials. If applicable, subject to the final sentence of this Section 3.1, within a commercially practicable time after written request from Sub-Recipient, NCDMM shall supply Sub-Recipient, if previously negotiated, Materials as set forth in the SOW. Sub-Recipient shall own all right, title and interest in and to any Materials, unless specifically identified to the contrary in writing. If NCDMM supplies Materials to Sub-Recipient, NCDMM shall provide Sub-Recipient with written documentation evidencing the value of such Materials (the "Valuation Price"). If the Parties disagree over the Valuation Price provided by NCDMM, after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

3.2 Information. If requested by Sub-Recipient to provide information in NCDMM's possession that is not otherwise accessible to Sub-Recipient and that is reasonable and necessary for Sub-Recipient's performance under this Agreement ("Information"), subject to the following sentence, NCDMM shall provide such Information (or an explanation of the legitimate reason for any refusal or delay, and if delay, a projected date by which such Information will be provided) within a commercially practicable period of time after receipt of Sub-Recipient's written request. Notwithstanding the foregoing, NCDMM may, in its sole discretion, refuse to provide any Information to Sub-Recipient, including Confidential Information, Inventions or any other Information NCDMM deems, in its sole discretion, to be proprietary or unnecessary for use in the Activities, provided that Sub-Recipient shall not be deemed in breach of this Agreement if the sole reason it cannot perform any Research or Development Activities is NCDMM's refusal to provide Information reasonable and necessary for the conduct of the Activities. NCDMM shall own all right, title and interest in and to any Information, unless specified to the contrary in writing.

3.3 Use of Materials and Information. Sub-Recipient shall (i) use the Materials and Information only for the purposes described in the SOW or such other purposes as NCDMM may approve in writing, (ii) restrict access to and use of the Materials and Information to the Principal Investigator and other researchers employed by, or who are agents of, Sub-Recipient or any permitted Sub-Recipients for whom such access and use is required to conduct the Activities

and who are bound by obligations of confidentiality at least as strict as those set forth in Section 6.

3.4 **Supply of Other Materials by Sub-Recipient.** All equipment and materials other than the Materials set forth in the SOW, required for the Activities shall be purchased and paid for by Sub-Recipient, unless otherwise agreed to in writing by NCDMM. Sub-Recipient shall own all right, title and interest in and to such equipment and materials purchased under this Section 3.4, unless specifically identified to the contrary in writing.

3.5 **Non-Exclusive License.** If, in its sole discretion, NCDMM determines that it is necessary for Sub-Recipient's conduct of the Activities, NCDMM may, by separate written agreement with Sub-Recipient, grant Sub-Recipient a non-exclusive, non-transferable and non-assignable right and license to use an Invention owned by, or licensed (with the right to sublicense) to, NCDMM, which license shall be limited in duration to the Term or such shorter period of time designated by NCDMM, limited in territory to use in a Facility, and limited in scope to use solely as needed to perform the Activities (or a part thereof as designated by NCDMM). For the avoidance of doubt, this Agreement does not grant any such license.

ARTICLE 4. BUDGET AND PAYMENT

4.1 **Compensation and Payment Terms.** NCDMM is entering into a Cost Reimbursement No Fee Agreement with the Sub-Recipient in accordance with the approval of AFRL found in Schedule B. In consideration for the satisfactory performance of the Activities by Sub-Recipient in accordance with this Agreement, subject to the provisions of this Section 4, NCDMM shall, within forty-five (45) to sixty (60) days of receipt of an invoice and any supporting documentation in accordance with Section 4.5, pay to Sub-Recipient the amounts for the invoiced incurred expenses. The Parties agree that NCDMM shall pay no Amounts or Expenses until it has been funded under the Prime Contract with respect to the Research and Development Activities. All payments hereunder shall be made in U.S. dollars, by check.

4.2 **Reimbursement of Expenses.** NCDMM shall reimburse Sub-Recipient for actual expenses incurred by Sub-Recipient for reasonable, necessary and verifiable:

4.2.1 Materials that are set forth in the SOW that NCDMM does not provide to Sub-Recipient;

4.2.2 Travel expenses incurred while performing the Activities in accordance with the Joint Travel Regulations as published by the Federal Government("Travel Expenses");

4.2.3 Other expenses expressly set forth in Schedule B (Budget) ("Other Approved Expenses");

4.2.4 Any other expenses specifically approved in advance in writing by NCDMM.

Collectively, the expenses referred to in Sections 4.2.1 through 4.2.4, are referred to herein as the "Expenses." Notwithstanding the foregoing, Sub-Recipient shall only be reimbursed for Expenses that are allowable costs under 2 Code of Federal Regulations 200, Subpart E. Sub-Recipient acknowledges and agrees that NCDMM shall not pay or reimburse Sub-Recipient for any Expenses incurred by, or purchases made by or on behalf of, Sub-Recipient other than those set forth in this Section 4.2.

4.3 **Total Obligation.** Unless otherwise agreed in writing by the Parties, the foregoing amounts represent NCDMM's total obligation for any and all Activities and any and all expenses incurred by or on behalf of Sub-Recipient. The total funds authorized by this agreement shall not exceed \$899,500 in federal funding and \$513,065 cost share.

4.4 Expense Changes. The Expenses are based on the scope of work and assumptions stated in the SOW. If NCDMM requests a change in the SOW which the Parties agree, in good faith, would result in an increase to Sub-Recipient of its cost of conducting the Activities, the Parties shall, in good faith, evaluate and discuss revised Expenses. Based on this evaluation and discussion, the Parties shall amend the SOW and Budget by written agreement. In the event the Parties cannot agree on a price change after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

4.5 Invoicing and Progress Reports.

4.5.1 Sub-Recipient shall invoice NCDMM once each month during the Term. Each invoice shall be accompanied by a written progress report, which shall describe the Activities Sub-Recipient has performed since the date of the last invoice and provide such other information as may be required by this Agreement or reasonably requested by NCDMM. Each monthly invoice shall include expenses and cost share for expenses incurred to NCDMM's reasonable satisfaction due Sub-Recipient since the date of the last invoice. For Expenses to be reimbursed pursuant to Section 4.2, Sub-Recipient shall include in the monthly invoice a summary report of the Expenses incurred since the date of the last invoice.

4.5.2 If NCDMM disputes any portion of an invoice, it shall pay the undisputed portion and shall provide Sub-Recipient with written notice of the disputed portion and its reasons therefor, and NCDMM shall not be obligated to pay interest on such disputed portion. In the event the Parties cannot agree on an invoice after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

4.5.3 **Closeout, Adjustment, Continuing Responsibilities and Reimbursement** – Final payment will not be made nor can this agreement be closed out until the sub-recipient delivers to NCDMM all the final closeout paperwork, disclosure of subject inventions, and the final project report. NCDMM may adjust downward the total obligation for the effort if this information is not provided.

4.6 Records Retention and Audit.

4.6.1 Records Retention. Sub-Recipient shall keep or cause to be kept accurate records or books of account in accordance with applicable generally accepted accounting principles that, in reasonable detail, fairly reflect the invoiced Fees and Expenses. Such books and records shall be maintained by Sub-Recipient for at least three (3) years following the end of the calendar year to which they pertain.

4.6.2 Audit. Upon the written request of the United States Government or NCDMM, Sub-Recipient shall permit a Government auditor, or a certified public accountant or a person possessing similar professional status and associated with an independent accounting firm reasonably acceptable to the Parties, to inspect during regular business hours all or any part of Sub-Recipient's records and books necessary to verify invoices related to Expenses and Cost Share.

4.7 Arbitration. In the event that a Party requests a Valuation Price determination in accordance with Section 3.1, a price change determination in accordance with Section 4.4, or an invoice determination in accordance with Section 4.5.2 to be submitted to an arbitrator, such determination shall be determined by a single arbitrator in accordance with the Commercial Arbitration Rules of the American Arbitration Association. The decision of the arbitrator shall be final and conclusive, absent manifest error. The Parties shall amend this Agreement if and as appropriate to implement the decision of the arbitrator.

ARTICLE 5. REPRESENTATIONS, WARRANTIES AND COVENANTS

5.1 Sub-Recipient represents and warrants to, and covenants with, NCDMM that:

5.1.1 Sub-Recipient shall perform the Activities in good scientific manner and in compliance with Applicable Law and shall use reasonable efforts to perform the Activities in accordance with the SOW.

5.1.2 Neither Sub-Recipient nor any of its principal officers or employees has been debarred or suspended by any governmental entity from conducting any Activities or is subject to any such debarment and Sub-Recipient will not use in any capacity, in connection with its obligations under this Agreement, including via any permitted Sub-Recipient, any person who has been so debarred.

5.1.3 Each Facility is and shall be during the Term in material compliance with all requirements of Applicable Law. There are, and during the Term shall be, no pending or uncorrected citations or adverse conditions noted in any governmental inspection of any Facility with respect to any Activities.

5.1.4 Sub-Recipient owns or has the legal right to use all patents, know-how and other intellectual property rights required for use in its conduct of the Activities, excluding those rights obtained from NCDMM under separate written agreement of the Parties or owned by or licensed to NCDMM pursuant to this Agreement or otherwise.

5.1.5 The Principal Investigator is, and at all times during the conduct of the Activities shall be, qualified by training and experience with appropriate expertise to conduct and supervise the Activities.

5.1.6 Sub-Recipient (for itself and each Facility) and the Principal Investigator have, and at all times during the conduct of the Activities shall have, all appropriate licenses, approvals, authorizations, registrations, permits, and certifications necessary to safely, adequately and lawfully perform the Activities. Sub-Recipient shall provide copies of all such licenses, approvals, authorizations, registrations, permits or certifications upon NCDMM's request.

Sub-Recipient shall advise NCDMM immediately if any of the foregoing in this Section 5.1 ceases to be true and correct.

5.2 Each Party represents and warrants to, and covenants with, the other Party that:

5.2.1 It has the requisite power and authority to execute, deliver and perform its obligations under this Agreement and all governmental and third-party approvals necessary for it to enter into and perform its obligations under this Agreement have been obtained, are in full force and effect and are final and non-appealable.

5.2.2 This Agreement is its legal, valid, and binding obligation, enforceable in accordance with its terms and conditions.

5.2.3 The execution and delivery of this Agreement and the transactions contemplated in it do not violate, conflict with, or constitute a default under the terms or provisions of any agreement or other instrument to which it is a party or by which it is bound, or any order, award, judgment, or decree to which it is a party or by which it is bound.

5.2.4 It is in compliance with all Applicable Law and it will comply with all Applicable Law during the Term.

A Party shall immediately notify the other Party if any of the foregoing in this Section 5.2 with respect to such first Party ceases to be true or correct.

5.3 Disclaimer. EXCEPT AS EXPRESSLY SET FORTH IN THIS SECTION 5, NCDMM MAKES NO REPRESENTATIONS AND GRANTS NO WARRANTIES, EXPRESS OR IMPLIED, EITHER IN FACT OR BY OPERATION OF LAW, BY STATUTE OR OTHERWISE, UNDER THIS AGREEMENT, AND NCDMM SPECIFICALLY DISCLAIMS ANY OTHER WARRANTIES, WHETHER WRITTEN OR ORAL, OR EXPRESS OR IMPLIED, INCLUDING ANY WARRANTY OF QUALITY, MERCHANTABILITY OR FITNESS FOR A PARTICULAR USE OR PURPOSE WITH RESPECT TO ANY MATERIALS (AS DEFINED IN SECTION 1.1.7) OR ANY WARRANTY AS TO THE NON-INFRINGEMENT OF ANY INTELLECTUAL PROPERTY RIGHTS OF THIRD PARTIES UNDER THIS AGREEMENT.

ARTICLE 6. CONFIDENTIALITY

6.1 In the normal and routine completion of this project, there may be some occasions where one Party discloses information that is proprietary and confidential to the disclosing Party. All such information which a disclosing Party believes to be proprietary, confidential, trade secret or the like undisclosed information shall be provided in writing or other permanent visual form (such as, for example, a physical prototype, physical sample, photograph, video or sound recording on any type of recording media) and designated as proprietary with an appropriate restrictive legend stamped or marked thereon at the time disclosure. Proprietary Information disclosed orally, must be identified as proprietary at the time of disclosure and summarized in writing as being proprietary within thirty (30) days of such disclosure.

6.2 The receiving Party agrees to exercise the same care and safeguards with respect to Proprietary Information disclosed by the disclosing Party as used to maintain the confidentiality of its own information of like character, but in no event less than a reasonable degree of care.

6.3 The obligations of confidentiality and non-disclosure set forth in this section 6 shall continue for a period of five (5) years from the termination or expiration of this Agreement or until the information ceases to fall under the definition of Proprietary Information.

ARTICLE 7. INTELLECTUAL PROPERTY

7.1 Background Intellectual Property. Each Sub-Recipient/Sub-Recipient/Member shall retain all rights to its Background Intellectual Property; and the decision to make available any such Background Intellectual Property for use in a Sub-Recipient/Member's sub award project shall be at the sole discretion of each Sub-Recipient/Member and in accordance with DOD regulations with respect to identification of all such Background Intellectual Property. No license or rights are granted to a Sub-Recipient/Member's Background Intellectual Property under this Agreement.

7.1.1 In the event that one Sub-Recipient/Member may require use of another Sub-Recipient/Member's Background Intellectual Property that has been disclosed by a Sub-Recipient/Member as part of the Consortium Research in order to successfully commercialize any CDIP then the Sub-Recipient/Members agree to discuss potential licensing terms and

conditions in a separate legally binding agreement between the Sub-Recipient/Members, separate from this Agreement. Sub-Recipient/Member(s) are not required to license any such originating Sub-Recipient/Member(s) Background Intellectual Property.

7.2 Consortium Developed IP shall be owned by the respective inventing or creating organizations, subject to any government rights and/or any pre-existing rights of any third party and subject to the following conditions:

7.2.1 If a Party solely or jointly creates CDIP, the Party must disclose the creation of such CDIP to its technology transfer office, licensing office or other similar department ("Party's TechTransfer Office"). A non-confidential summary of the CDIP disclosed to the Party's TechTransfer Office shall be sent to NCDMM as soon as practicable so that NCDMM can maintain a list of CDIP. Sub-Recipient/Members owning CDIP shall grant upon request to NCDMM and Sub-Recipient/Members-In-Good-Standing at the time of creation a limited, non-exclusive, royalty-free license to use the CDIP for the Sub-Recipient/Member's internal procedures, research or development purposes (but not to make, use, or sell products or external processes for commercial purposes, with the exception of licenses granted pursuant to Section 7.2.3). Such licenses shall be granted to interested Sub-Recipient/Members upon request in a separate legally binding mutually agreeable license agreement between the Sub-Recipient/Members. Payment of patent expenses may be required of Parties granted non-exclusive, royalty-free commercial licenses by universities and other non-profit institutions. Such licenses for Sub-Recipient/Members shall be without the right to grant sublicenses to third parties, except for any Sub-Recipient/Member-designated agents, contractors and non-employee students ("Permitted Third Parties") performing work for the benefit of such Sub-Recipient/Member. Under these circumstances the Sub-Recipient/Member is responsible for having any and all appropriate written agreements with such Permitted Third Parties to enable Sub-Recipient/Member's compliance with this Agreement and is responsible for such parties' use of the CDIP in the same manner Sub-Recipient/Member is responsible for its own use of such CDIP (e.g., violation of the license parameters set forth in this section by a Sub-Recipient/Member's Permitted Third Parties shall be considered a breach of this Agreement by Sub-Recipient/Member).

7.2.2 Sub-Recipient/Members are strongly encouraged to seek legal protection in the form of patents as soon as is reasonably possible following disclosure of all CDIP to NCDMM. Protection of a Sub-Recipient/Member's solely developed CDIP shall be done at Sub-Recipient/Member's own expense and through use of their respective Party's TechTransfer Office. Each Sub-Recipient/Member agrees to notify in writing NCDMM in a timely manner of all such actions in which legal protection is or has been sought so that NCDMM can enter such information in its invention disclosure database. With respect to jointly developed CDIP the relevant Sub-Recipient/Members agree to negotiate a separate legally binding agreement encompassing those terms and conditions to be used to govern the manner in which jointly developed CDIP will be owned, administered, protected, and licensed. NCDMM will be notified in writing in a timely manner of the existence of these agreements between Sub-Recipient/Members and NCDMM shall maintain pertinent information in its invention disclosure database. In the event that a single Sub-Recipient/Member for solely developed CDIP, or all Sub-Recipient/Members with an ownership right for jointly developed CDIP choose not to seek legal protection and thereby elect not to file a patent application on any

CDIP, then Sub-Recipient/Member(s) agree to notify NCDMM in writing of its intent and must report any pending publication or presentation to NCDMM at the time of this notification. NCDMM may negotiate to obtain such protection at its own expense where Sub-Recipient/Member(s) choose not to seek legal protection. Ownership of CDIP shall remain with the originating Sub-Recipient/Member(s).

7.2.3 It is anticipated that one of the outcomes of an active IP licensing and commercialization plan is the generation of royalty income by a respective Sub-Recipient/Member. It is acknowledged that Sub-Recipient/Members of academic, government, and industry sectors will manage the disposition and reporting requirements of all royalties received in accordance with their institution's existing policies, through their Party's TechTransfer Office. To the extent it may legally do so, Sub-Recipient/Members owning CDIP shall grant NCDMM and Platinum Sub-Recipient/Members-In-Good-Standing a limited, non-exclusive, royalty-free license to use the CDIP for commercial purposes and Gold Sub-Recipient/Members-In-Good-Standing shall be granted such licenses at a fair market value royalty rate. Such licenses shall be granted to interested Sub-Recipient/Members upon request in a separate legally binding mutually agreeable agreement between the Sub-Recipient/Members. Such licenses for Sub-Recipient/Members shall be without the right to grant sublicenses to third parties, except for any Sub-Recipient/Member-designated agents, contractors and non-employee students performing work for the benefit of such Sub-Recipient/Member, provided, however, the Sub-Recipient/Member is responsible for having any and all appropriate written agreements with such parties to enable Sub-Recipient/Member's compliance with this Agreement and is responsible for such parties' use of the CDIP in the same manner Sub-Recipient/Member is responsible for its own use of such CDIP (e.g., violation of the license parameters set forth in this section by a Sub-Recipient/Member's contractor shall be considered a breach of this Agreement by Sub-Recipient/Member). In addition, sublicensing shall be permitted to the licensee's direct customers.

7.3 Licenses granted under this Section 7 shall be subject to these additional terms:

Any licenses granted to a Sub-Recipient/Member through a separate document as specified in this Section 7 shall become a royalty bearing license at fair market value for such a Sub-Recipient/Member who is designated a Withdrawn Sub-Recipient/Member.

It is understood that the United States Government (through any of its agencies or otherwise) may provide use of its facilities or equipment and / or may provide funds for Consortium Research. As a result, this Agreement, any and all rights and obligations of the Sub-Recipient/Members to any CDIP resulting from use of any United States Government's facilities, equipment or funds are subject to any and all applicable rights of the United States Government.

Notwithstanding anything to the contrary in this Agreement, certain laws, regulations and/or policies may prevent and/or limit certain Sub-Recipient/Members' ability to offer royalty-bearing licenses to CDIP that has previously been licensed by such Sub-Recipient/Members on a royalty-free basis. Therefore, the ability to charge royalties to Gold Sub-Recipient/Members, Silver Sub-Recipient/Members, and/or third parties is subject to the granting Sub-Recipient/Member's ability to do so in light of then-existing contractual obligations, legal and regulatory requirements, and policies of the granting Sub-Recipient/Member.

ARTICLE 8. INDEMNIFICATION AND INSURANCE

8.1 **Indemnification by Sub-Recipient.** Sub-Recipient agrees to defend, indemnify and hold harmless NCDMM and its Affiliates and their respective directors, officers, employees and agents from and against any and all costs (including reasonable legal fees), damages, expenses and losses (collectively, “Losses”) arising from any third party suit, claim, demand, assessment, action or proceeding (collectively, “Claims”), in any manner caused by, resulting from or arising out of any misrepresentation, negligence or breach of this Agreement on the part of Sub-Recipient, any of Sub-Recipient’s Affiliates or any permitted Sub-Recipients, or any of Sub-Recipient’s or its Affiliates’ or permitted Sub-Recipients’ directors, officers, employees or agents, including Losses arising out of the breach or inaccuracy of any of Sub-Recipient’s representations, warranties, or covenants under Section 5 of this Agreement, except to the extent any such Loss is attributable to any breach by NCDMM of this Agreement, or any negligence, wilful misconduct or unreasonable inaction by NCDMM, any of NCDMM’s Affiliates, or any or NCDMM’s or its Affiliates’ directors, officers, employees or agents.

8.2 **Indemnification by NCDMM.** NCDMM agrees to defend, indemnify and hold harmless Sub-Recipient and its Affiliates and their respective directors, officers, employees and agents from and against any and all Losses arising from any third party Claims, in any manner caused by, resulting from or arising out of any misrepresentation, negligence or breach of this Agreement on the part of NCDMM, or any of NCDMM’s directors, officers, employees or agents, including any Losses arising out of any breach or misrepresentation made by NCDMM under Section 6, except to the extent any such Loss is attributable to any breach by Sub-Recipient of this Agreement, or any negligence, wilful misconduct or inaction by Sub-Recipient, any of Sub-Recipient’s Affiliates or any permitted Sub-Recipients, or any of Sub-Recipient’s or its Affiliates’ or any permitted Sub-Recipients’ directors, officers, employees or agents.

8.3 **Indemnification Procedures.** In the event that either Party (an “Indemnified Party”) seeks indemnification from the other Party (the “Indemnifying Party”) under the terms of this Article 8 with respect to any claim made by a third party, it shall inform the Indemnifying Party of the claim as soon as reasonably practicable after it receives notice thereof, shall permit the Indemnifying Party, at the Indemnifying Party’s cost, to assume direction and control of the defense of the claim, and shall co-operate as requested (at the expense of the Indemnifying Party), in the defense of the claim. After notice to the Indemnified Party of the Indemnifying Party’s election to assume the defense of such claim, the Indemnifying Party shall be liable to the Indemnified Party only for such legal or other expenses subsequently incurred by the Indemnified Party in connection with the defense thereof at the request of the Indemnifying Party. The Indemnifying Party shall not settle or otherwise compromise any claim or suit without the prior written consent of the Indemnified Party, which shall not be unreasonably withheld. As to those third party claims with respect to the which the Indemnifying Party does not elect to assume control of the defense, the Indemnified Party will afford the Indemnifying Party an opportunity to participate in such defense, at the Indemnifying Party’s own cost and expense, and will not consent to the entry of any judgment, settle or otherwise dispose of any of the same without the consent of the Indemnifying Party, which shall not be unreasonably withheld.

8.4 Limitation of Liability and Claims. NCDMM shall not be liable to Sub-Recipient hereunder for any indirect, incidental, special, punitive or consequential damages of any kind, including lost profits or loss of goodwill or otherwise; and in no event shall the collective, aggregate liability of NCDMM and its Affiliates and its and their respective directors, officers, employees and agents under this Agreement exceed the amount of compensation actually received by Sub-Recipient from NCDMM pursuant to this Agreement. No action, regardless of form, arising out of or in any way connected with this Agreement or the Activities, may be brought by Sub-Recipient more than one (1) year after the cause of action accrued.

ARTICLE 9. PUBLICATIONS

9.1 The Parties agree and expect that results of Consortium Research associated with this agreement shall be published or otherwise made publicly available and that Parties engaged in Consortium Research shall be permitted to present at symposia, national or regional professional meetings and to publish in journals, theses or dissertations, or by other means of their own choosing, the results of their research, provided that nothing will be done which could bar the availability of patent protection with respect to CDIP of a Sub-Recipient/Member or America Makes (NAMII) invention or which would disclose Proprietary Information of any Sub-Recipient/Member or of America Makes (NAMII) or disclose information in violation of the applicable U.S. laws and regulations (e.g., the International Traffic in Arms Regulations (“ITAR”) and the Export Administration Regulations (“EAR”) that govern the export of specific technical data and technologies, including software, prototypes and other intellectual property, to foreign countries and foreign nationals (“Export Control Laws”).

9.1.1 A Party will not make a public disclosure without a review of the full text of the proposed publication, presentation or other form of public disclosure by the Sub-Recipient/Member(s) involved, the America Makes (NAMII) Director and Government PM as described below. The Sub-Recipient/Member(s) involved, the America Makes (NAMII) Director and Government PM shall be provided a copy of the proposed public disclosure at least sixty-five (65) days in advance of the submission of such proposed public disclosure and shall have two (2) weeks after receipt of said proposed disclosure to respond in writing to the submitting Party to identify Proprietary Information and/or to identify any potentially patentable CDIP and/or to identify any CDIP in which the submitting Party does not have an ownership interest. A submitting Party agrees to remove any identified Proprietary Information, potentially patentable CDIP and/or CDIP in which the submitting Party does not have an ownership prior to public disclosure (or, for potentially patentable CDIP in which the submitting Party does have an ownership interest, delay public disclosure for a period of sixty (60) days from the date of the response).

9.1.2 Notwithstanding anything to the contrary above, student theses and dissertations shall be subject to a separate review and comment process wherein the student shall submit such student thesis or dissertation in draft form at least sixty (60) days in advance of the date of their final defense in order to afford an opportunity to identify Proprietary Information and/or identify any potentially patentable CDIP and/or any CDIP in which the Party’s student does not have an ownership interest.

9.1.3 America Makes (NAMII) and/or America Makes (NAMII) Sub-Recipient/Members may negotiate and implement a more restrictive public disclosure agreement than defined in paragraphs 9.1.1 and 9.1.2 for a specific America Makes (NAMII) project based on the need for extended non-disclosure of CDIP by the project participants.

9.1.4 An acknowledgment of funding and a disclaimer shall appear in the publication of any material, whether copyrighted or not, resulting from an America Makes (NAMII) project incorporating U.S. Government funds granted in support of the America Makes (NAMII) Consortium. The acknowledgement shall read:

“This material is based on research sponsored by Air Force Research Laboratory under agreement number FA8650-20-2-5700. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon.”

The disclaimer shall read:

“The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of Air Force Research Laboratory or the U.S. Government.”

ARTICLE 10. TERM AND TERMINATION

10.1 Term. The term of this Agreement shall begin on the date hereof and shall continue until December 23, 2026 (the “Term”). Subject to termination in accordance with Section 10.2, the Parties agree to discuss in good faith prior to the end of the Term amending this Agreement (including this Section 10.1 and Schedules A, B, C, and D (if applicable)) or entering into a separate agreement governing the Parties’ relationship with respect to the general substance of this Agreement, subject to mutual agreement, for one or more years following the end of the Term.

10.2 Termination.

10.2.1 Either Party shall have the right to terminate this Agreement, by giving written notice to the other Party, upon material breach by the other Party of any material provision of this Agreement, including any covenant, representation or warranty contained in this Agreement or failure to perform any material portion of the Activities in accordance with the SOW, provided that such breach shall have continued for a period of thirty (30) days after the non-breaching Party has delivered written notice of said breach to the breaching Party.

10.2.2 This Agreement may be terminated at any time by written agreement of the Parties.

10.2.3 NCDMM shall have the right to terminate this Agreement for any or no reason upon thirty (30) days prior written notice to Sub-Recipient.

10.2.4 NCDMM shall have the right to terminate this Agreement immediately upon termination of the Prime Agreement (in whole or part) or upon any amendment or other change to the Prime Agreement that materially affects the need for or value of (each in NCDMM’s reasonable sole discretion) all or a portion of the Activities hereunder.

10.2.5 Sub-Recipient shall have the right to terminate this Agreement upon thirty (30) days written notice if circumstances beyond its reasonable control preclude continuation of the Activities as described in the SOW.

10.3 Rights and Duties Upon Termination or Expiration. Upon termination or expiration (without a further agreement related to the substance hereof entered into as contemplated in

Section 10.1) of this Agreement, (i) Sub-Recipient shall cease all work on the Activities and turn over to NCDMM all documentation prepared by (or on behalf of, as the case may be) Sub-Recipient for NCDMM during the Activities (whether in written or electronic form) and (ii) NCDMM shall, in accordance with Article 4 pay Sub-Recipient. Termination or expiration of this Agreement shall be without prejudice to any rights that shall have accrued to the benefit of a Party prior to such termination or expiration.

10.4 Survival. The respective rights and obligations set forth in the previous Articles and any other provisions required to interpret and enforce the Parties' rights and obligations under this Agreement shall indefinitely survive the expiration or termination of this Agreement to the extent expressly set forth therein or if no survival term is expressly set forth therein to the extent necessary to preserve these rights and obligations.

ARTICLE 11. MISCELLANEOUS

11.1 No Commercialization. Sub-Recipient agrees that it shall not commercialize or, except as expressly permitted under the terms of this Agreement, otherwise Exploit any Materials, Information or Inventions owned by or licensed to NCDMM.

11.2 Equitable Relief. Sub-Recipient acknowledges and agrees that the restrictions set forth in Articles 6 and 7 and Sections 8.1 and 8.2 of this Agreement are reasonable and necessary to protect the legitimate interests of NCDMM and that NCDMM would not have entered into this Agreement in the absence of such restrictions, and that any violation or threatened violation of any provision of Article 6 or 7 or Section 8.1 or 8.2 will result in irreparable injury to NCDMM. Sub-Recipient also acknowledges and agrees that in the event of a violation or threatened violation of any provision of Article 6 and 7 or Section 8.1 or 8.2, NCDMM shall be entitled to preliminary and permanent injunctive relief, without the necessity of proving irreparable injury or actual damages and without the necessity of having to post a bond. The rights provided in the immediately preceding sentence shall be cumulative and in addition to any other rights or remedies that may be available to NCDMM. Nothing in this Section 11.2 is intended, or should be construed, to limit NCDMM's right to preliminary and permanent injunctive relief or any other remedy for a breach of any other provision of this Agreement.

11.3 Publication. Sub-Recipient, including its Affiliates and any permitted Sub-Recipients and its or their respective employees or agents, has no right to publish the NCDMM name or logo, except as expressly consented to in writing by NCDMM.

11.4 Waiver. The failure of a Party to enforce any breach or provision of this Agreement shall not constitute a continuing waiver of such breach or provision and, subject to Section 8.4, such Party may at any time thereafter act upon or enforce such breach or provision of this Agreement. Any waiver of breach executed by either Party shall affect only the specific breach and shall not operate as a waiver of any subsequent or preceding breach. To be effective any waiver must be in writing.

11.5 Assignment and Foreign Participation. Sub-Recipient shall not sell, transfer, assign, pledge or otherwise dispose of, whether voluntarily, involuntarily, by operation of law or otherwise, this Agreement or any of its rights or obligations hereunder without the prior written consent of NCDMM, provided that in the event such consent and approvals, if applicable, are granted, the proposed assignee shall agree in writing, in a form reasonably satisfactory to

NCDMM, to perform all of Sub-Recipient's obligations hereunder. Any attempt to assign, transfer, subcontract or delegate any portion of this Agreement or Sub-Recipient's rights or obligations hereunder in violation of this Section 11.5 shall be null and void. Subject to the terms hereof, this Agreement shall be binding upon and inure to the benefit of the successors and permitted assigns of the respective Parties hereto.

Any foreign person or other foreign participation conducting work under this effort in or outside of the United States must be approved prior to any of the work being completed. Requests for approval should be submitted to the cognizant NCDMM project manager managing the effort. The request will be submitted to the Government program office for final review and approval. No work can be completed outside of the United States without this approval.

11.6 Severability. If any provision of this Agreement is held to be illegal, invalid or unenforceable under any present or future law, and if the rights or obligations of any Party under this Agreement will not be materially and adversely affected thereby, (a) such provision shall be fully severable, (b) this Agreement shall be construed and enforced as if such illegal, invalid or unenforceable provision had never comprised a part hereof, (c) the remaining provisions of this Agreement shall remain in full force and effect and shall not be affected by the illegal, invalid or unenforceable provision or by its severance herefrom and (d) in lieu of such illegal, invalid or unenforceable provision, there shall be added automatically as a part of this Agreement, a legal, valid and enforceable provision as similar in terms to such illegal, invalid or unenforceable provision as may be possible and reasonably acceptable to the Parties.

11.7 Notices. Any notice, request or other communication required to be given pursuant to the provisions of this Agreement shall be in writing and shall be deemed to be given when delivered in person, by facsimile or by courier (return receipt requested) or five (5) days after being deposited in the United States mail, postage prepaid, certified, return receipt requested to the Parties addressed as follows:

If to NCDMM, to:	If to Sub-Recipient, to:
Gary R. Fleegle, President & CEO	Barbara Ewing, Contracts
699 Scalp Avenue	241 W Federal Street
Johnstown, PA 15904-1619	Youngstown, OH 44503
Tel: (724) 539-1361	Tel: (330) 746-5003
gary.fleegle@ncdmm.org	bewing@ybi.org

Either Party may change its contact person, address or telephone or facsimile number by giving the other Party written notice, delivered in accordance with this Section 11.7.

11.8 Construction. Except where the context otherwise requires, wherever used, the singular shall include the plural, the plural the singular, the use of any gender shall be applicable to all genders and the word "or" is used in the inclusive sense ("and/or"). When this Agreement refers to a number of days, unless otherwise specified as business days, that reference is to calendar days. The captions of this Agreement are for convenience of reference only and in no way define, describe, extend or limit the scope or intent of this Agreement or the intent of any provision contained in this Agreement. The term "including" as used herein shall mean including, without limiting the generality of any description preceding such term. A reference to a Schedule, Article, Section or clause is a reference to a Schedule, Article, Section or clause

of this Agreement, and the terms “hereof,” “herein,” and other like terms refer to this Agreement as a whole, including any attachments hereto. The language of this Agreement shall be deemed to be the language mutually chosen by the Parties and no rule of strict construction shall be applied in favor of or against either Party hereto.

11.9 Governing Law. This Agreement and the rights and obligations of the Parties shall be governed by and construed under the laws of the Commonwealth of Pennsylvania, excluding any conflicts or choice of law rule or principle that might otherwise refer construction or interpretation of this Agreement to the substantive law of another jurisdiction. Code of Federal Regulation parts and clauses shall be governed by Federal law, including the Federal common law of agreements.

11.10 Parties Independent. In making and performing this Agreement, the Parties act and shall act at all times as independent entities and nothing contained in this Agreement shall be construed or implied to create an agency, partnership or employer and employee relationship between NCDMM and Sub-Recipient. Except as expressly provided herein, at no time shall either Party make commitments or incur any charges or expenses for or in the name of the other Party.

11.11 Conflicts. Except as otherwise required by Applicable Law, to the extent that any provision of this Agreement conflicts with a provision of a Schedule, the provision of this Agreement shall govern.

11.12 Cumulative Remedies. Unless expressly set forth herein to the contrary, all remedies set forth herein are cumulative and are in addition to any and all remedies provided either Party at law or in equity.

11.13 Entire Agreement; Amendment. This Agreement and the previously signed America Makes (NAMII) Sub-Recipient/Membership agreement contain the entire understanding of the Parties with respect to the subject matter hereof. Any amendment or supplement to this Agreement shall be effective only if in writing signed by each Party.

11.14 Counterparts. This Agreement may be executed in any number of counterparts, each of which shall be an original as against the Party whose signature appears thereon, but all of which taken together shall constitute but one and the same instrument.

IN WITNESS WHEREOF, each Party has caused this Sub-Recipient Agreement to be signed by its duly authorized officer as of the date first above written.

NCDMM	[Sub-Recipient]
By: _____ (Signature)	By: _____ (Signature)
Name: _____ (Printed Name)	Name: _____ (Printed Name)

Title: _____ Date: _____	Title: _____ Date: _____
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SCHEDULE A

**Youngstown Business Incubator (YBI) proposal titled Additive Manufacturing for Metalcasting:
The Last Tactical Mile, dated June 14, 2024, and all associated Exhibits are hereby incorporated
by reference and in their entirety.**

Additive Manufacturing for Metalcasting: The Last Tactical Mile

Submission Entity: The Youngstown Business Incubator (YBI)

For America Makes Open Project Call 2024

Solicitation Response to Impact Part A Topic 2

Project Team

Youngstown Business Incubator
University of Northern Iowa Industry 4.0 Center
Ohio State University Center for Design and Manufacturing Excellence
University of Tennessee Knoxville
Boeing
HA International
Humtown Additive
Defense & Energy Systems, LLC
M&P Gravity Works, LLC
3D Systems
Matthews Additive

Principal Investigator

Stephanie Gaffney
Vice President of Advanced Manufacturing Programs, Youngstown Business Incubator

YBI Duns: 9459100730000
YBI Cage Code: 5EAR9

14 JUNE 2024

Additive Manufacturing for Metalcasting: The Last Tactical Mile

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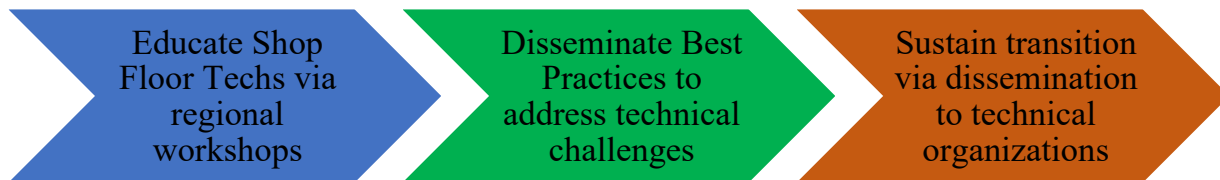
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1.0 Executive Summary

The Youngstown Business Incubator (YBI) and its long standing casting technology partners including the University of Northern Iowa's Industry 4.0 Center (UNI), the Ohio State University's

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Center for Design and Manufacturing Excellence (OSU CDME) and the University of Tennessee Knoxville, along with industry partners including Humtown Products, Boeing, HA International, Matthews Additive Technology and 3D Systems have prepared a comprehensive team to accelerate the transition of additive manufacturing (AM) and related enabling technologies to the domestic casting base to facilitate rapid part replacement in the legacy fleet and support new platform employment. The “Additive Manufacturing for Metalcasting (AM4MC) - The Last Tactical Mile” project offers three courses of action to further penetrate casting houses and increase technology application in real time.



Our team will develop a compelling curriculum and execute **eight** regional workshops to directly engage shop floor technicians and young engineers who do not have the ability to attend national venues to move technical competence at foundries beyond basic understandings of AM to actual integration based on applications. Part of this engagement includes educating company owners of second and third generation casting houses on the business value to move them from blocking technology (CAPEX) and workforce investments to embracing innovation. Our team will leverage each of its research partners to develop content which address technical barriers to adoption and include that content in our planned courses, both the face to face and virtual events. We will further sustain innovation via presentation of best practices and portions of the curriculum at venues to the supply base including America Makes events, AFS, AMC, ICI and Steel Founder events including national and AM specific venues and leadership forums.

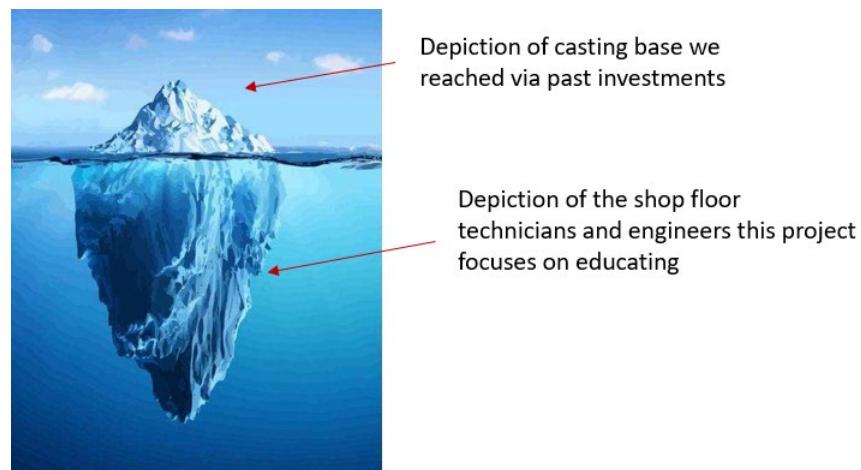


Figure 1. Target audience: Shop floor technical staff not yet reached

Our objective is simple: Show value to America Makes members including OEMs, Tier 1 and 2 entities and defense program offices via accelerated development of their small and medium manufacturing (SMM) supply chain. Target an audience left behind, i.e., technicians and young engineers on the shop floor who drive innovation in the small and medium manufacturing

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enterprises and ensure their leadership understands the value proposition therein. To execute this plan our team requests \$900,000 for the 27-month period of performance in addition to our cost share contribution of \$450,000 providing a total value of over \$1.35M. Key project deliverables are listed in table 1.

Table 1: Project Deliverables and Outcomes

Deliverable	Target Audience	Outcomes
Develop & deliver an 8-hour seminar and present it at 8 regional events-AM for Metalcasting Course	500 shop floor technicians, engineers and managers	Accelerated uptake of AM principles based on an application driven perspective
Develop & deliver two hours of Business Value Content	Foundry owners and senior managers & investors	C Suite level support for advanced technology investment and workforce development
Sustainable delivery of project content including curriculum and best practices to the domestic casting base	Metal casting industry at large including forums hosted by America Makes, the AFS, Steel Founders, ICI, AMC, etc.	Effective leveraging of professional technical societies to augment America Makes members down to the shop floor



Figure 2. Program Team

2.0 Technology Approach and Methodology

Our uniquely experienced team proposes an innovative project focusing on leveraging previous work, but with a new focus on a yet to be reached audience found deep in the metal casting base that were overlooked by past efforts. This approach will enable a widespread increase in capacity and capability and maximize the investment of resources expended in this initiative.

2.1 Problem Statement and America Makes Relevance

The Department of Defense has invested significant resources through various entities such as America Makes to accelerate industry uptake of compelling technologies. For example, the team submitting this proposal has led three previous America Makes projects focusing on accelerating

Additive Manufacturing for Metalcasting: The Last Tactical Mile

utilization of additive manufacturing for casting manufacturing and at least one DLA project with similar objectives. Despite these previous investments, OEMs and Tier One entities are still challenged to procure legacy systems or new platform castings in a timely manner. Some of our OEM partners experience lead times exceeding 12-24 months for complex aerospace castings such as gear boxes and constant speed drives. This impacts the availability of platforms to meet combatant command requirements. Our approach is well suited for informing America Makes members who attend TRX and MMX events such as OEMs, Tier One entities, printer OEMs, engineering services and material entities of the value proposition of directly engaging shop floor technicians and young engineers on the benefits of binder jetting and material extrusion technologies for casting cost, schedule and part performance gains. Even America Makes non-profit members, including academia, labs and government partners, will directly benefit from this project's technical content dissemination to their workforce such as the defense organic base, including artisans and engineers.

2.2 Deep Knowledge of AM Technology in the Domestic Casting Base

The collective DOD investment impact on the domestic supply chain has been significant, judging by the interactions at the recent AFS 2024 Cast Expo held in Milwaukee in April of this year. At that forum, 14 of the 77 technical presentations occurring during the three-day event were focused on the utilization of AM technologies for sand casting. In past Cast Expo venues, AM related presentations were mostly offered by research organizations. Of those 14 presentations, 10 were made by members of our project team. Unlike previous venues where academia led the charge, at this most recent event, industry led AM application discussions and companies included Mercury Marine, HA International, Grede, 3D Systems, Matthews, etc. Moreover, the AFS Additive Manufacturing Division is now one of the most popular of AFS working groups. Early adopters such as Southern Cast Products, Danko Arlington and Humtown Products have demonstrated the value proposition of cost, schedule and part consolidation to dozens of defense OEMs who reap the benefits in overall platform sustainment. An additional data point includes large format sand printer procurement, which was less than five systems deployed in 2014 when our team started these transition projects. Our team now conservatively estimates over 110 printers in operation in the U.S. casting base.

This begs the question of why invest more precious defense resources for additional uptake of AM for sand casting if our observations demonstrate such positive trend lines? Part of the answer to this question requires a review of past success stories. The Youngstown Business Incubator (YBI) tracked the business success of four regional companies pursuing additive manufacturing as their core competitive advantage. The companies included Additive Engineering Solutions (AES), Juggerbot3D, Humtown Products and Freshmade3D.

Average time for these companies to achieve significant revenue generation was 6.5 years to penetrate small portions of the North American manufacturing market for their core business offerings. During those years, all of these companies benefited from non-dilutive capital, including funds from regional entities or national partners such as America Makes or SBIR/STTR funds, to help them move to profitability and impact the defense industrial base in a positive trend. **Government investment must now be focused on the last tactical mile** and persistent to achieve measurable improvement by our domestic base to OEMs.

Additive Manufacturing for Metalcasting: The Last Tactical Mile

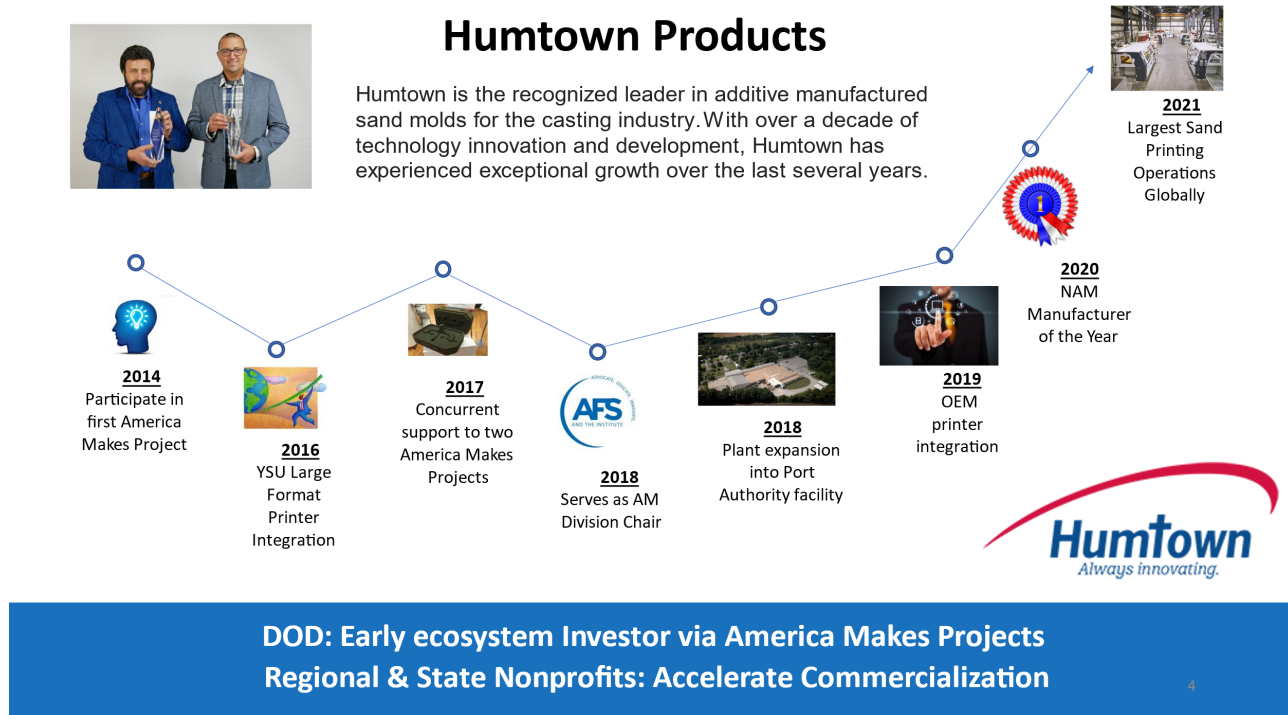


Figure 3. Humtown AM for Metal Casting Growth Timeline

2.3 Project Work Scope, Objectives and Overcoming Barriers: Achieving Desired Results

Scope & Objectives. Our team's AM4MC Last Tactical Mile three-pronged approach has a simple work scope: improve the supply chain through organic growth of AM capability and decidedly compress timelines for industry uptake of additive manufacturing for metalcasting. To ensure project success, our team has identified several objectives including:

- Establishment of essential industry knowledge around AM utilization for metalcasting
- Enable a macro strategy of dissemination to shop floor technicians via well planned and attended regional events across the U.S.
- Proliferate widespread success and employment of AM and enabling technologies to achieve increased capacity as measured by Tier 1s and OEMs
- Sustain the momentum via continued sharing of project outcomes to the America Makes audience and also existing technical organizations' forums, workshops and expos.

Overcoming Barriers: The Last Tactical Mile's Competitive Edge

A 2022 industry survey pointed out several challenges to adoption. Foremost were the elements of printer and material costs and workforce knowledge. Regarding cost, our project team will inform the audience on the emergence of lower cost printers and cost-efficient materials and binders systems in our curriculum. Moreover, our aggressive one-day class is filled with key educational elements to build the confidence of shop floor technicians on when and how to use additive manufacturing and other enabling technologies including AI, robotics & sensors.

Competitive Edge: Other proposers are offering curriculum, which has previously been "sold" to individual foundries. Or, the offerings occur as part of a national event such as Cast Expo. Our

Additive Manufacturing for Metalcasting: The Last Tactical Mile

project trumps those efforts by going one step further to address two **key challenges** we have found through many years of industry engagement. First, we must go to them via one day workshops near their locations to get the workforce education numbers required for significant uptake of AM. Second, despite education of the young engineers and technical staff at foundries, **foundry owners, managers and investors often fail to understand the financial and business benefits of additive manufacturing**, workforce education, robotic post processing and in situ process monitoring collectively have on the company's bottom line. We are differentiating this project proposal from competitors by offering Foundry 4.0 seminars at leadership focused events. These concise one-to-two-hour presentations target C suite executives attending forums such as the AFS, Steel Founders and ICI annual leadership conferences. The University of Iowa's Center for Business Growth & Innovation will provide the content and delivery, presenting material in an experienced fashion to gain leadership support. Details of this work are listed in Task 1.1.

2.4 AM4MC Last Tactical Mile Project Tasks

Task 1: Educate via Development and Dissemination of AM for Metalcasting Curriculum

Primary Focus:

- Address educational needs across the **entire** metal casting workforce.

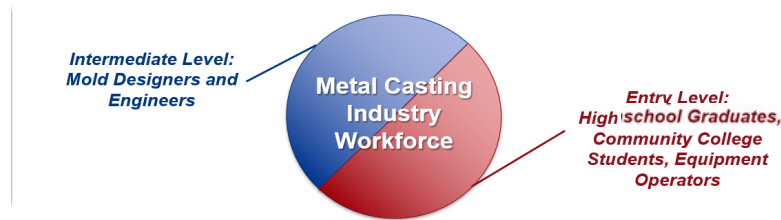


Figure 4. Target Audience: Shop floor technical workforce

Our team will leverage a renowned team of industry leaders and academia to prepare and present a one-day course to educate foundry technical staff and supporting industry workers on the key elements of AM application for sand casting. The presenters will be offering cost share to develop the curriculum and then be paid to travel to designated sites and deliver the curriculum throughout the United States at regional forums organized by our team and delivered at sponsor foundries or existing industry professional chapters. The course content will include topics listed in table 2.

Table 2: 1 Day AM Seminar Curriculum Potential Content

Topic	Content Focus	Presenter
Innovations in sand printing, cleaning, processing and post processing of printed cores	Demonstrate to the audience best practices for these challenging castings	Nathaniel Bryant and Jerry Thiel, University of Northern Iowa Industry 4.0 Center;
Printed Mold & Core Design Fundamentals to speed production and post processing of high value castings	Educate shop floor technicians on the utilization of Binder Jetting for appropriate applications to accomplish improved quality, part consolidation, lead times	David Rittmeyer, Matthews International; Dr. Jason Walker, OSU CDME
Extrusion Technology for Production Patterns and Core	Understanding the application of extrusion technology and	Marshall Miller, Applications Engineer-Pellet Extrusion

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Equipment and Combined Additive Processes with Binder Jetted Sand	compared advantages over standard subtractive methods and when it makes sense to combine additive processes	Technology for the foundry, 3D Systems
Buying versus Making: All you need to know about printed tooling production	Present best practices on printed tooling manufacturing and machine challenges.	Brandon Lamoncha, Humtown Additive, Jerry Thiel, UNI
Materials and Applications to address surface finish and other technical challenges	Primer on how various materials including sand, ceramics, binder systems, coatings. impact casting production	Kelly Kerns, HA International; Dr. Dustin Gilmer, University of Tenn, Knoxville
Emerging Technologies and their Impact on AM for metal casting	Impacts of low-cost printers, binder systems, scanning technology, direct metal printing, challenging geometries, alloys, etc.	Technical Panel: All

Planned Dissemination Targets for Regional Workshops: Our comprehensive Industry and Academic Team has an aggressive strategy of starting workshops within 90 days of contract signing to ensure we present a minimum of eight one-day seminars during the 24-month period of performance. **These workshops will not only attract industry, but also focus on attracting defense acquisition professionals (artisans and engineers) via this curriculum to meet their continuous learning requirements.** Our team will leverage its experience in a past America Makes project to ensure we attract 40-75 defense technical personnel to augment the industry attendees at our Oklahoma City venue at Rose State Community College, a short drive from Tinker AFB. Tinker and Rock Island already possess large format binder jet printers and thus they will directly benefit from this workshop. The one-day workshop curriculum will align with Table 3 lists regional targets for the seminars.

Table 3: Initial list of regional one day classes across the U.S.

Event	Location	Target Date
Mid Atlantic Workshop	Donsco Foundry, Wrightsville, PA	Fall 2024
SE Region Workshop	Huntsville, AL	Spring 2025
Tinker AFB Workshop	Rose State Community College, Oklahoma City, OK	FEB 2025
Upper Midwest Workshop	Milwaukee, WI	Summer 2025
Mid OH Workshop	OSU CDME, Columbus, OH	Fall 2025
Michigan/Indiana Workshop	Detroit, MI	FEB 2026
Central Region Workshop	Waterloo, IA	Spring 2026
Western Region Workshop	Colorado School of Mines, Golden, CO	Summer 2026

After the execution of three regional chapter events our expert team will adjust the curriculum and content delivery as appropriate. Our workshop team represents the best experts in the field regarding adoption of additive manufacturing for metal casting. Shown in figure 5, this group presented the majority of the AM related content at the 2024 Cast Expo annual conference and they represent the senior leadership of the AFS AM Division team.

Additive Manufacturing for Metalcasting: The Last Tactical Mile

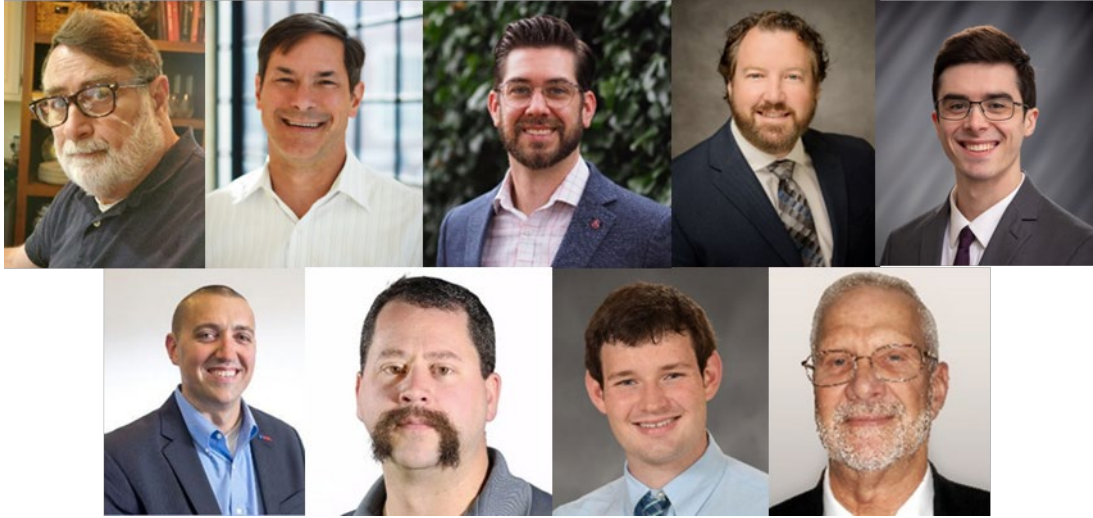


Figure 5. Renowned AM Workshop Presenters

Task 1.1: Educate Foundry Leaders to enable Technology Investment

A second component of our regional events is focused on owners. As stated in the executive summary, a barrier to AM uptake is leadership decision making regarding the value proposition of technologies that foundry engineers request owners invest in to increase operational efficiency. With guidance from the president of Wellman Dynamics, our team from the University of Northern Iowa's Center for Business Growth and Innovation, led by Todd Hutcheson, a former senior manager at Rockwell International and Collins Aerospace, and an experienced defense product line manager, will develop curriculum via cost share and then leverage grant resources to engage the domestic foundry base leadership at existing forums or during our regional events. Ideally this line of effort will provide foundry owners decision with the tools by which they can measure AM and related capital investments, as well as metrics on workforce investments. Specific foundry leadership engagement objectives include:

1. *Objective 1:* Establish a value proposition framework: This allows companies to improve their understanding of benefits, impact, and trade-off analysis, demonstrating the value of technology in improving manufacturing floor costs (including prototype and product turn-time, quality/scrap, environmental costs, safety, talent attraction/retention, labor productivity and cash flow among many others).
2. *Objective 2:* Enhance decision-making: This allows casting & foundry companies to use the results obtained from the framework established in Objective 1 to determine when it is of the greatest business value to deploy AM capabilities for certain forged products.
3. *Objective 3:* Target specific needs: This enables an improved understanding of the value that underlying technologies like AM and IOT, etc. as well as improved workforce development, availability, and technology augmentation, bring to a business. Accomplishing this objective builds support from shop floor technicians and provides detailed information to leadership to make appropriate CAPEX value and risk mitigation decisions.
4. *Objective 4:* Provide knowledge of the value framework to technical experts: This will educate technical experts on how to identify and prioritize qualification requirements, risks, and validate qualification test matrices.

Additive Manufacturing for Metalcasting: The Last Tactical Mile

To accomplish these objectives, the UNI Center for Business Growth and Innovation will attend five industry and government stakeholder conferences which traditionally are attended by foundry leadership teams. Planned conferences include:

Table 4: List of Leadership Conferences to engage foundry owners

Foundry Leadership Event	Event Date
2024 AFS Foundry Leadership Summit	SEP 2024
2025 NFFS Industry Executive Conference	FEB 2025
2025 ICI Business & Leadership Development Conference	APR 2025
AFS Cast Expo Global Conference 2025	APR 2025
ICI AM4IC	JUN 2025

Task 2: Facilitate Further Sustainable Transition and Pervasive Impacts to Industry

Our project team subject matter experts will sustain this project's innovation drive by offering the curriculum and casting trial observations to all of the domestic technical organizations such as AFS, Steel Founders, AMC, ICI, etc. via presentations and engagement to their members to enable broad workforce and business leader development of our critical casting supply chain. Each of these organizations hold annual conferences in addition to tech specific conferences throughout the United States. Our team has presented at these venues numerous times and thus we know how to gain access, what to present and to whom in the industry we will be presenting. Based on America Makes membership rules and Department of Defense guidance, our project content will be offered to these technical organizations for long standing integration into their knowledge centers. Completion of this task will facilitate deeper penetration of the supply chain.

Table 5: Project dissemination (in addition to CORE/Storefront Content Availability).

Event	Location	Date	Audience
America Makes TRX	Los Angeles	Fall 2024	Members & DOD
ICI AM Event	TBD	Summer 2025	Domestic SMMs
Cast Expo 2025	Atlanta	APR 2025	Casting Industrial Base
AFS & ICI Leadership Events	TBD	Summer 2025	Foundry Owners
AFS AM Conference	Indianapolis	Feb 2026	Domestic SMMs
AFS AM Expo	TBD	Spring 2026	Industry at large

2.5 Key Performance Parameters and AM Tech Roadmap Alignment

AM Technical Roadmap Alignment

This project directly aligns with past roadmapping alignment efforts including a 2016 NIST funded roadmap in addition to the most recent America Makes document titled AM-Technology Roadmap for Casting-Forging Report dated OCT 2023.

Regarding the America Makes roadmap, our Task 1 webinar content and our Task 2 focused efforts address the areas highlighted on Figure 4, p. 15 of the 2023 roadmap, depicted in the figure below.

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1	Modeling and simulation tools to improve the design process
2	Develop tools/guides to assist with technology selection and design
3	Printing sand molds and cores
4	Reverse engineering tools to shorten design lead times
5	Printing ceramic molds and cores
6	Printing patterns
7	Use AM to produce a traditionally cast part

Figure 6. Highlights of AM-Technology Roadmap for Casting-Forging Report

Key Performance Parameters (KPPs): During the America Makes Casting and Forging Roadmapping workshops, industry identified several investment areas to focus research funding. The resultant work for the cast part demonstrations should be baselined against traditionally manufactured cores and machined tooling. Industry identified KPPs include the following:

Table 6: Key performance parameters with baseline and improvement criteria.

Task	KPP	Baseline	Target	Rationale
1 YBI	Number of People Foundry Professional Attendees at AFS AM Seminars	No current practice.	Average 25 per event in-person. 200 total.	Increasing the number of people impacted through in-person seminars enhances knowledge dissemination and industry engagement.
1 YBI	Number of Views on YouTube of Recorded Workshops & Technical Material	No current practice.	1,500 views.	Increasing online views of recorded seminars and technical material expands the reach and accessibility of the content, benefiting a wider audience.
2 Team	Number of companies reached at AFS, ICI, Professional Society	Limited data	500 companies reached during our cost shared presentations at multiple sand and investment casting venues throughout the	Provides a comprehensive engagement of the domestic casting supply chain including engineering and material and equipment providers necessary to support technology adoption.

Additive Manufacturing for Metalcasting: The Last Tactical Mile

	Workshops		27 month period of performance.	
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Table 7: Additional Key performance parameters with baseline and improvement criteria.

KPP	Baseline	Improvement Criteria
Improved casting process repeatability for an aerospace part with long pipe cores	Compare current/traditional core making and assembly to AM printed core integration	20% reduction in scrap rate
Reduced tooling manufacturing costs	Compared to traditional core production	15% reduction in costs
Improved Shake Out Methods	Compare traditional methods for long thin pipe core shakeout to this new process methods	25% reduction in core shakeout issues
Address surface finish issues with printed cores	Compared to current state of the technology	50% increase in casting acceptance

Specific project outputs will be shared with membership via the digital storefront (CORE) and presentation at America Makes TRX events. Our industry partners at Boeing, Wellman Dynamics, and Humtown Products will assess our project outputs as they continue manufacturing and utilization of printed molds and cores for specific defense parts. The industry team will provide our group real time feedback on the value proposition so that we can modify or add content to our Task 1 and 2 efforts.

3.0 Data Management Plan

The project team will be compiling data obtained from activities funded under this grant. The plan will ensure compliance and data availability to the appropriate entities within the project team including authorized America Makes members and our defense project management team.

1. Types of data, samples, and materials to be produced during the project.

This project will produce several types of data including Task 2 body of information compiled as cast part SolidWorks files and printer focused STL files for cores and molds in addition to Word documents outlining Task 1 course content, numerous PowerPoint briefings and reports mostly as PDFs. The data captures will be characterized in the following manner: observational, experimental, reference and derived

2. Standards to be used for data and metadata format and content.

The data will be stored in the following formats: STL files for solid geometry parts, PDF, Word and PPT and video content as required. The types of metadata that will be included

Additive Manufacturing for Metalcasting: The Last Tactical Mile

are statistical and descriptive, which can be leveraged by America Makes technical membership.

3. Methods and policies for providing access and enabling sharing.

Data will be stored on the America Makes CORE site to ensure appropriate management, updates and security. The YBI management team will ensure all team members comply with this plan.

4. Provisions for re-use, re-distribution, and the production of derivatives.

Data will be made available on or after the following date: SEP 2024 upon project start. YBI will work with the designated America Makes program manager to establish an acceptable folder and how schema will be identified, uploaded, and updated throughout the 27-month period of performance. Team members will be briefed on this data management plan at the start of work meeting. All reproductions of project information must be approved by the YBI and America Makes project management authorities to ensure preservation of data and archiving.

4.0 Technology Dissemination to America Makes Members, Impact to OSD ManTech and Likelihood of Impact to DoD Broader Supply Chain Partners.

Dissemination of project best practices will occur at all America Makes TRX venues throughout the period of performance. In addition, the key project team members will attend America Makes MMXs and also use virtual forums such as the Maintenance & Sustainment Working Group and lunch time virtual venues to reach a broad set of America Makes members including OEMs, Tier 1, 2 and 3 entities.

Pervasive Impacts: Given our aggressive engagement of the U.S. industrial base, OSD ManTech will enjoy direct feedback from key casting providers such as Wellman Dynamics and OEMs like Sikorsky on how the project is increasing the capabilities of the casting base. OSD ManTech can inform PEOs and acquisition leadership and congressional leaders how this project is quantifiably improving the U.S. industrial base via accelerated technology transition of compelling additive manufacturing technology and other enabling technology to improve acquisition of critical cast parts needed to sustain legacy systems and field new systems.

5.0 Project Management Plan

Project Management Approach

YBI will manage the 27-month project. YBI will host bi-weekly and monthly technical meetings to ensure the tasks are performed in the allotted time. YBI has experience and capabilities in managing large technical efforts in support of America Makes and State of Ohio efforts, including several AFRN and ARC projects. The management team will be comprised of YBI and Defense & Energy Systems key personnel. At the technical level, UNI MCC will assist YBI in assessing combined university efforts. We will rotate quarterly meetings to on-site forums at each of our university partners to facilitate interim and final project reviews.



Additive Manufacturing for Metalcasting: The Last Tactical Mile

Data management will follow the prescribed data management plans discussed at our Start of Work meeting, which will be held within 30 days of project contract. At that forum, YBI will define user permissions and roles in order to ensure protection of sensitive data and any proprietary information. Project leads and technical personnel from each team member will receive data access rights based on their individual and specific roles. America Makes and AFRL project managers will also receive access to any data storage per the contract guidelines.

Moreover, YBI will engage OEM directed casting houses immediately upon project award to facilitate scheduling of the eight regional workshops. Humtown Products will assist YBI in the contracting of foundry design work, printed molds, and foundry contracting for specific demonstrations. This engagement is critical in ensuring best practices in casting demonstrations can be performed during the short period of performance. YBI will leverage knowledge management tools such as Air Table to track and share work. Weekly calls are required to keep the collective team focused during the five-month research and demonstration phase. Reporting and data capture will be enabled through the America Makes CORE portal to ensure timely dissemination and availability of findings to membership. YBI has demonstrated past performance of several past America Makes efforts, including three focused on AM technology transition to the U.S. domestic base.

No.	Milestone	Quarter								
		1	2	3	4	5	6	7	8	9
1	Develop and evolve workshop curriculum									
2	Execute Regional Workshops									
3	Present Findings and Progress at TRX and Technical Forums									
4	Report Analysis and Documentation									
5	Final Reporting									

Figure 7: Project work plan and milestones.

The project work plan includes detailed analysis and follow-on demonstrations of the project technical deliverables and then presentation to America Makes Members and casting industrial base SMEs. Specific work plan activities for the effort include the following milestones:

Defense Need	• Robust supply chain, capable of making complex lightweight castings for both the rapidly expanding commercial aerospace market and for manned and unmanned defense systems platforms. • High quality parts produced using repeatable processes which minimize human error in the manufacturing process. • Rapid tooling, supporting both sustainment and new systems development.
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Exhibit I. Identification of Background Intellectual Property

Project Title	Additive Manufacturing for Metal Casting-The Last Tactical Mile
Organization	Youngstown Business Incubator
Principal Investigator	Stephanie Gaffney

List all known background intellectual property to be used in the conduct of this project or for which access may be required to implement project results:

Inventor / Owner	Title	Patent or Disclosure I.D.
none		

☒ If controlled by a project participant, I understand that a “good faith” commitment to enter into negotiations for a license of this background intellectual property to America Makes or the project partners may be required.

OR

☐ I am unaware of any background intellectual property to be used in the conduct of this project or that may be required for implementation of project results.

Intellectual Property Rights Policy & Confidentiality Statement

As the Principal Investigator at _____ (Member) participating in this project, I agree to accept and abide by the Intellectual Property Rights Requirements of the America Makes Membership Agreement, as approved by the Governance Board, dated _____. I understand that I may be the recipient from time to time of information of a confidential and proprietary nature belonging to another America Makes participating organization. I have read and explicitly agree to abide by the provisions Section 5 of the America Makes Membership Agreement with respect to proprietary information.

I further agree to assist the project participants in their obligation of implementing America Makes' intellectual property requirements for funded projects. I will do this by encouraging the timely submission of invention disclosures by project participants to their appropriate Intellectual Property Office, clearly identifying such disclosures as relating to this project, and by providing any supporting documentation and information that may be requested from time to time for the purpose of filing patent applications under America Makes and/or the Inventing Organization(s).

Principal Investigator

Signature		
Printed Name	Stephanie Gaffney	DATE: 08/07/2024

Exhibit II. Multiple Submissions Summary

America Makes recognizes that projects may be submitted to multiple sources of funding.

America Makes must be informed if other funding is secured and will work with the Principle Investigator (PI) to modify this project scope, as appropriate.

List all planned or submitted requests for additional funding of work in this project area from sources other than America Makes.

Date Submitted or Planned Submittal Date	Organization	Decision Date
none		

Principal Investigator

Signature	
Printed Name	Stephanie Gaffney
Date	08/07/2024

Exhibit III. Letters of Commitment

(If the sub-recipient is already involved in the Phase 1 work, then a general statement of additional commitment in terms of the funding required, the cost share contributed, and the total value to this proposal is all that is necessary.)

Exhibit IV. Acknowledgement of Consortium Developed Intellectual Property and Disclosure of Planned Publications

x As the Principal Investigator at Youngstown Business Incubator (Member), I agree that results from this project will be considered consortium-developed intellectual property (CDIP) and will be shared amongst the membership according to the consortium-developed intellectual property (CDIP) structure in the Membership Agreement, with exceptions made for any ITAR/Export Control information.

☐ As the Principal Investigator at _____ (Member) participating in a funded Project, I project the following publications of the results from this project. In addition, I agree that all publications will be approved through the AFRL Technical POC 45 days prior to public dissemination.

Summary of Planned Publications from Project Team
None known at this time

Principal Investigator

Signature		
Printed Name	Stephanie Gaffney	DATE: 08/7/2024

Exhibit IV. Identification of Consortium Developed Intellectual Property

List all Consortium Developed Intellectual Property that will be delivered at the end of this project:

List of project CDIP/Deliverable Artifacts and Completion Dates (examples including, but not limited to best practices, cost-benefit analyses, COTS Capability, benchmark datasets, demonstration study, equipment specs, hardware, material & process specs, performance evaluations, procedures/guidelines/tutorials, process windows, quality assurance methods, reference test coupons, software/algorithms, and validated models.) Note that this list should align with your technical deliverable list. Add pages to the Exhibit V table as needed.

Organization Name / Responsible Party	Consortium Developed Intellectual Property	Detail Specific Tangible Artifacts to be Delivered to Membership	Expected Delivery Date to Membership	Associated Background Intellectual Property	ITAR Restricted (yes/no)
YBI	Kick-Off Meeting	Timeline, deliverables and team member expectations in PPT slide	NLT one month after award	N/A	no
YBI	Planned project deliverables list with corresponding (existing/new) Technology Roadmap Requirements	Spreadsheet outlining deliverables, timing, delivered to Core, America Makes (w/o Core), and type (slides, document, written report, data, test report results, etc.)	NLT 4 months after subrecipient award	N/A	no
YBI	Technical progress reports	Quarterly report describing technical progress for the quarter; report per template Quarterly Report Template - America Makes.doc	Monthly, or more as needed	N/A	no
YBI	Financial and man hour reports	Expenditure reports as generated by YBI	monthly	N/A	no
YBI	Data management plan including data persistence plan	Document that includes definition and	Document that includes definition and	N/A	no

YBI	Virtual technical update meeting slides presented at project update call, or virtual meetings between project team, America Makes, and gov. PMs and advisors	handling of all digital data and physical artifacts and retention Presentation of technical progress for pertinent month; report per Monthly Update Presentation Template (America Makes).ppt	handling of all digital data and physical artifacts and retention Monthly, or more as needed	N/A	no
YBI	Presentations at public forums such as America Makes TRX, MMX, or industry recognized conferences	Minimum of 5 national briefings to America Makes members	NLT 24 months after subrecipient award	N/A	no
YBI	Workshops at regional chapter events	Minimum of 8 one day (8 hour) regional workshops across the US Metrics report of attendees including (company, position, and employment category) Digital copy of PowerPoint slides presented at workshops Digital copy of all recorded presentations from the workshops	NLT 24 months after the subrecipient award	N/A	no
YBI	Presentations at Foundry Events such as AFS, NFFS, ICI or other industry recognized conferences	Minimum of 5 two-hour presentations across the US Metrics report of corporations in attendance	NLT 24 months after subrecipient award	N/A	no

		Digital copy of PowerPoint slides presented at conferences. Digital copy of all recorded presentations from the workshops			
	Draft Final Project Report	Written draft report using Final Report Template (America Makes).doc	NLT 22 months after subrecipient award	N/A	no
YBI	Key Performance parameter report with data collected and presented in graphical form	Included in the final report	NLT 24 months after subrecipient award	N/A	No
	Written Final Project Report (minimum one round of revision completed with PM/NCDMM prior to final submission). At a minimum report will outline: critical process parameters, lessons learned, standards used and any deviations from those standards, target market/applications, team members and roles, motivations and reasoning for technical approach, KPPs, summary of tasking, methods, standards and specifications used, deviations to those standards or specifications, outcomes/results, summary and observations and analysis of data gathered, summarized allowable datasets, how data substantiated successful	Written final report using Final Report Template (America Makes).doc	NLT 24 months after subrecipient award	N/A	no

	outcome of this effort, and future opportunities for research and development realized through this effort.			
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Principal Investigator

Signature	<i>Stephanie Gaffney</i>	
Printed Name	Stephanie Gaffney	Date: 10-1-2024

Exhibit VI. Proposal Cost Summary

Project Title	Additive Manufacturing for Metal Casting-The Last Tactical Mile
Organization	Youngstown Business Incubator
Address	241 West Federal Street Youngstown, Ohio 44503
Principal Investigator	Stephanie Gaffney

Overall Requested Funding

America Makes	\$899,500	Duration	(in months)
Cost Share	\$513,065		27
Project Total	\$1,412,565		

Lead Organization

Name	Youngstown Business Incubator	Phone	330-503-0561
Address	241 W Federal Street Youngstown, Ohio 44503	Fax	
Cost Share (\$)	213,037	Email	sgaffney@ybi.org

Partner

Name	University of Northern Iowa	Phone	319-273-3217
Address	213 East Bartlett Hall	Fax	
Cost Share (\$)	33,062	Email	Maggie.heretakis@uni.edu

Partner

Name	The University of Tennessee Knoxville	Phone	931-393-7297
Address	411 B.H. Goethert Parkway Tullahoma, TN 37388	Fax	
Cost Share (\$)	15,735	Email	dgilmer@vols.utk.edu

Partner

Name	The Ohio State University	Phone	
Address	1080 Carmack Road Columbus, Ohio 43210	Fax	
Cost Share (\$)	20,272	Email	Walker.1762@osu.edu

Partner

Name	Defense and Energy Systems	Phone	330-719-8545
Address	1802 Paulin Road Poland Ohio 44514	Fax	
Cost Share (\$)	30,000	Email	Rich@ dne.systems

Partner

Name	Humtown Products	Phone	330-482-5555
Address	8301 Market Street Boardman, Ohio 44512	Fax	
Cost Share (\$)	30,000	Email	brandon@humtown.com

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Partner

Name	3D systems	Phone	803-326-3900
Address	333 Three D systems Circle Rock Hill, SC 29730	Fax	
Cost Share (\$)	30,000	Email	Marshall.miller@3dsystems.com

Partner

Name	Matthews International	Phone	260-273-2690
Address	1315 West Liberty Ave Pittsburgh, PA 15226	Fax	
Cost Share (\$)	30,000	Email	drittmeyer@matw.com

Partner

Name	Precision Casting Technologies	Phone	319-415-0379
Address	3227 Hwy D65 Dysart, IA 52224	Fax	
Cost Share (\$)	30,000	Email	thiel@uni.edu

Partner

Name	HA International	Phone	630-575-5700
Address	630 Oakmont Lane Westmount, IL 60559	Fax	
Cost Share (\$)	30,000	Email	Kelley.kerns@ha-group.com

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America Makes RFP

Partner

Name	M&P Gravity Works	Phone	234-759-0071
Address	PO Box 70 North Lima, Ohio 44452	Fax	
Cost Share (\$)	30,000	Email	Krodgers126@gmail.com

SCHEDULE B

BUDGET

Team Member Name: Youngstown Business Incubator

Proposal Number: Impact 2.0 topic 2

		GRAND TOTAL
	REQUESTED AMERICA MAKES FEDERAL FUNDING	
	LABOR	212,326
	TRAVEL	0
	SUBCONTRACT	0
	MATERIALS	0
	EQUIPMENT	0
	CONSULTANT	605402
	ODC's	0
	INDIRECTS	81772.76
	TOTAL REQUESTED AMERICA MAKES FEDERAL FUNDING	899,500
	COST SHARE	
	LABOR	0
	TRAVEL	0
	SUBCONTRACT	0
	MATERIALS	0
	EQUIPMENT	0
	CONSULTANT	279,522
	ODC's	0
	UNRECOVERED INDIRECTS	\$ 233,543.12
	TOTAL PROPOSED COST SHARE	513,065
	TOTAL PROJECT EFFORT	1,412,565

SCHEDULE C

APPLICABLE FEDERAL REGULATION AND PRIME AGREEMENT FLOW-DOWN

1.00 Administrative Requirements and Order of Precedence (Mar 2015)

(a) This award is governed by the guidance in 2 Code of Federal Regulations (CFR) part 200, “Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards” as modified and supplemented by the Department of Defense’s (DoD) interim implementation found at 2 CFR part 1103, “Interim Grants and Cooperative Agreements Implementation of guidance in 2 CFR part 200” (79 FR 76047, December 19, 2014), all of which are incorporated herein by reference. Provisions of Chapter I, Subchapter C of Title 32, CFR, “DoD Grant and Agreement Regulations,” other than part 33, continue to be in effect and are incorporated herein by reference, with applicability as stated in those provisions.

(b) In the event of a conflict between the terms of this agreement and other government documents the conflict shall be resolved by giving precedence in descending order as follows:

- (1) Federal statutes
- (2) Federal regulations
- (3) 2 CFR part 200, as modified and supplemented by DoD’s Interim implementation found in 2 CFR part 1103
- (4) Award-specific terms and conditions
- (5) Attachments to this award, if any
- (6) In case of disagreement with any requirements of this award, the recipient shall contact the NCDMM in order to resolve the issue. The recipient shall not access any costs to the award or accept payments until the issue is resolved.

1.01 Trafficking in Persons (March 2015)

This award is subject to the requirements of section 106 (g) of the Trafficking Victims Protection Act of 2000 (22 U.S.C. 7104, as implemented by 2 CFR 175)

You as the sub-recipient and your employees may not—

- (a) Engage in severe forms of trafficking in persons during the period of time that the award is in effect; or
 - (ii) Procure a commercial sex act during the period of time that the award is in effect; or
 - (iii) Use forced labor in the performance of the award or subawards under the award.

1.02 Title to Property – Identified in Budget (March 2015)

Title to personal property acquired with agreement funds shall vest in the sub-recipient upon acquisition, except that supplies shall be managed in accordance with 2 CFR 200 Section 314. Title to real property shall vest in the sub-recipient subject to conditions contained in 2 CFR 200 Section 311. The sub-recipient shall dispose of real property in accordance with 2 CFR 200 Section 311.

1.03 Property System (March 2015)

Sub-recipient's property system shall meet the standards as set forth in 2 CFR 200 Section 310-316.

1.04 Cost Principles (March 2015)

Sub-recipient(s) shall comply with the cost principles as contained in 2 CFR 200, Subpart E, Cost Principles. Cost principles for determining allowability of costs applicable to lower tier, cost type contracts or awards under this agreement shall be determined by type of entity receiving the award.

1.05 Standards for Financial Management (March 2015)

Sub-recipient(s) financial management system shall comply with the standards identified in 2 CFR 200 Section 302.

1.06 Audit Requirements/Retention and Access to Records (March 2015)

Sub-recipient shall comply with the audit requirements of 2 CFR 200, Subpart F Audit Requirements and shall comply with the requirements appropriate for the type of entity receiving the award. Sub-recipient's financial records, supporting documents, statistical records, and all other records pertinent to an award shall be retained and access to them permitted in accordance with 2 CFR 200 Section 336.

1.07 Cost Sharing (August 2001)

Sub-recipient's contributions may count as cost sharing only to the extent that they comply with 2 CFR 200(200.306)

1.08 Export Control (March 2015) (TAILORED)

(a) Access to the technology developed under this agreement by foreign firms, institutions or individuals shall be controlled by the Sub-recipient under applicable U.S. export control laws.

(b) The sub-recipient shall receive NCDMM and grants officer approval before assigning or granting access to any work, equipment, or technical data generated or delivered under this agreement to foreign persons or their representatives. The notification shall include the name and country of origin of the foreign person or representative, the specific work, equipment, or data to which the person will have access.

1.09 Inventions (March 2015)

(a) The clause entitled Standard Patent Rights, (37 CFR 401.14(a)) is hereby incorporated by reference and is modified as follows: replace the word "recipient" with "Sub-recipient"; replace the words "agency," "Federal Agency" and "funding Federal Agency" with "NCDMM"; replace the word "contract" with "agreement"; delete paragraphs (g)(2), (g)(3) and paragraph (g)(1) first sentence to read "The recipient will include this clause for experimental, developmental or research work to be performed by a Sub-recipient." Paragraph (1), Communications, point of contact on matters relating to this clause will be the servicing Staff Judge Advocate's office.

(b) Interim or final Invention Reports 1) listing subject invention(s) and stating that all subject inventions have been disclosed, or 2) stating that there are no such inventions, shall be sent to the NCDMM. Please include in the subject line of the e-mail the contract number followed by the words "Invention Reporting." The Sub-recipient shall file Invention (Patent) Reports on the DO Form 882, Report of Inventions and Subcontracts, as of the close of each performance year and at the end of the term for this agreement. Annual reports are due 60 days after the end of each year of performance and final reports are due 60 days after the expiration of the final performance period. Negative reports are also required annually.

(c) The DD Form 882 may also be used for the notification any lower tier award for experimental, developmental or research work which contain a "Patent Rights" clause

(d) This provision also constitutes the request for the following information for any subject invention for which the recipient has retained ownership: 1) the filing date, 2) serial number and title, 3) a copy of the patent application and 4) patent number and issue date. Submittal shall be to the NCDMM.

1.10 Data Rights (March 2015)

(a) All rights and title to data, as defined in 48 CFR 27.401, generated under this agreement shall vest in the sub-recipient.

(b) The sub-recipient hereby grants to the U.S. Government a royalty free, world-wide, nonexclusive, irrevocable license to use, modify, reproduce, release, perform, display or disclose any data for Government purposes.

- “Government purpose includes competitive procurement, but not the rights to use, modify, reproduce, release, perform, display, or disclose technical data for commercial purposes or authorize others to do so”

(c) The sub-recipient is responsible for affixing appropriate markings indicating rights on all data delivered under the agreement. The Government will have unlimited rights in all data delivered without markings.

(d) The sub-recipient shall include this article, suitably modified to identify the parties, in all lower tier contracts and awards, regardless of tier, for experimental, developmental, or research work.

1.11 Publishing Project Results (March 2015)

(a) Publications. The Sub-recipient(s) is expected to publish or otherwise make publicly available the results of the work conducted under this subaward. One copy of all publications resulting from the project shall be forwarded to the NCDMM as it becomes available.

(b) An acknowledgment of awarding agency's support shall appear in the publication of any material, whether copyrighted or not. The acknowledgement shall read:

"This material is based on research sponsored by Air Force Research Laboratory under agreement number FA8650-20-2-5700. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon."

(c) The Sub-recipient is responsible for assuring that every publication of material based on or developed under this project contains the following disclaimer:

"The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of Air Force Research Laboratory or the U.S. Government."

1.12 Reporting Subawards and Executive Compensation (March 2015)

(a) Reporting of first-tier subawards.

(1) Applicability. Unless you are exempt as provided in paragraph d. of this award term, you must report each action that obligates \$25,000 or more in Federal funds that does not include Recovery funds (as defined in section 1512(a)(2) of the American Recovery and Reinvestment Act of 2009, Pub. L. 111-5) for a subaward to an entity (see definitions in paragraph e. of this award term).

(2) Where and when to report.

(i) You must report each obligating action described in paragraph a.1. of this award term to <http://www.fsrs.gov>.

(ii) For subaward information, report no later than the end of the month following the month in which the obligation was made. (For example, if the obligation was made on November 7, 2010, the obligation must be reported by no later than December 31, 2010.)

(3) What to report. You must report the information about each obligating action that the submission instructions posted at <http://www.fsrs.gov> specify.

(b) Reporting Total Compensation of Recipient Executives.

(1) Applicability and what to report. You must report total compensation for each of your five most highly compensated executives for the preceding completed fiscal year, if-

(i) the total Federal funding authorized to date under this award is \$25,000 or more; (ii)

in the preceding fiscal year, you received-

(A) 80 percent or more of your annual gross revenues from Federal procurement

contracts (and subcontracts) and Federal financial assistance subject to the

Transparency Act, as defined at 2 CFR

170.320 (and subawards); and

(B) \$25,000,000 or more in annual gross revenues from Federal procurement contracts (and subcontracts) and Federal financial assistance subject to the Transparency Act, as defined at 2 CFR 170.320 (and subawards); and

(iii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/execomp.htm>.)

(2) Where and when to report. You must report executive total compensation described in paragraph b.1. of this award term:

(i) As part of your registration profile at <http://www.ccr.gov>.

(ii) By the end of the month following the month in which this award is made, and annually thereafter.

(c) Reporting of Total Compensation of Subrecipient Executives.

(1) Applicability and what to report. Unless you are exempt as provided in paragraph d. of this award term, for each first-tier subrecipient under this award, you shall report the names and total compensation of each of the subrecipient's five most highly compensated executives for the subrecipient's preceding completed fiscal year, if-

(i) in the subrecipient's preceding fiscal year, the subrecipient received

(A) 80 percent or more of its annual gross revenues from Federal procurement contracts (and subcontracts) and Federal financial assistance subject to the Transparency Act, as defined at 2 CFR 170.320 (and subawards); and

(B) \$25,000,000 or more in annual gross revenues from Federal procurement contracts (and subcontracts), and Federal financial assistance subject to the Transparency Act (and subawards); and

(ii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/excomp.htm>.)

(2) Where and when to report. You must report subrecipient executive total compensation described in paragraph c.1. of this award term:

(i) To the recipient.

(ii) By the end of the month following the month during which you make the subaward. For example, if a subaward is obligated on any date during the month of October of a given year (i.e., between October 1 and 31), you must report any required compensation information of the subrecipient by November 30 of that year.

(d) Exemptions: If, in the previous tax year, you had gross income, from all sources, under \$300,000, you are exempt from the requirements to report:

(1) Subawards, and

(2) The total compensation of the five most highly compensated executives of any subrecipient. (e)

Definitions. For purposes of this award term:

(1) Entity means all of the following, as defined in 2 CFR part 25:

(i) A Governmental organization, which is a State, local government, or Indian tribe;

(ii) A foreign public entity;

(iii) A domestic or foreign nonprofit organization; (iv)

A domestic or foreign for-profit organization;

(v) A Federal agency, but only as a subrecipient under an award or subaward to a non-Federal entity. (2)

Executive means officers, managing partners, or any other employees in management positions. (3)

Subaward:

(i) This term means a legal instrument to provide support for the performance of any portion of the substantive project or program for which you received this award and that you as the recipient award to an eligible subrecipient.

(ii) A subaward may be provided through any legal agreement, including an agreement that you or a subrecipient considers a contract. ·

(4) Subrecipient means an entity that:

- (i) Receives a subaward from you (the recipient) under this award; and
- (ii) Is accountable to you for the use of the Federal funds provided by the subaward.

(5) Total compensation means the cash and noncash dollar value earned by the executive during the recipient's or subrecipient's preceding fiscal year and includes the following (for more information see 17 CFR 229.402(c)(2)):

- (i) Salary and bonus.
- (ii) Awards of stock, stock options, and stock appreciation rights. Use the dollar amount recognized for financial statement reporting purposes with respect to the fiscal year in accordance with the Statement of Financial Accounting Standards No. 123 (Revised 2004) (FAS 123R), Shared Based Payments.
- (iii) Earnings for services under non-equity incentive plans. This does not include group life, health, hospitalization or medical reimbursement plans that do not discriminate in favor of executives, and are available generally to all salaried employees.
- (iv) Change in pension value. This is the change in present value of defined benefit and actuarial pension plans.
- (v) Above-market earnings on deferred compensation which is not tax-qualified.
- (vi) Other compensation, if the aggregate value of all such other compensation (e.g. severance, termination payments, value of life insurance paid on behalf of the employee, perquisites or property) for the executive exceeds \$10,000.

1.13 Disclosure or Information (March 2015) (Tailored)

(a) The Sub-recipient shall not release to anyone outside the Sub-recipient's organization any unclassified information, regardless of medium (e.g., film, tape, document, media announcements, etc.), pertaining to U.S. Government Agency-Driven Projects unless-

- (1) The NCDMM Agreements Officer has given prior written approval; or
- (2) The information is otherwise in the public domain before the date of release.

(b) Requests for approval shall identify the specific information to be released, the medium to be used, and the purpose for the release. The Sub-recipient shall submit its request to the NCDMM Agreements Officer at least 65 days before the proposed date for release through NCDMM.

(c) The Sub-recipient agrees to include a similar requirement in each sub-agreement under this agreement. Sub-recipients shall submit requests for authorization to release through the NCDMM to the Agreements Officer.

1.14 Procurement System (March 2015)

The sub-recipient's procurement system shall comply with the standards contained in 2 CFR 200 Section 317-326.

1.15 Military Recruiting on Campus (March 2015)

As a condition for receipt of funds available to the Department of Defense (DoD) under this award, the Sub-recipient agrees that it is not an institution of higher education (as defined in 32 CFR 216) that has a policy or practice that prohibits or, in effect, prevents ROTC or military recruiting on campus, as described in DFARS 252.209-7005(b). If the Sub-recipient is determined, using the procedures in 32 CFR part 216, to be such an institution of higher education during the period of performance of this agreement, and therefore to be in breach of this clause, the NCDMM will cease all payments of DoD funds under this agreement and all other DoD grants and cooperative agreements to the sub-recipient, and it may suspend or terminate such grants and agreements unilaterally for material failure to comply with the terms and conditions of award.

1.16 Assurances (March 2015)

(a) By signing or accepting funds under the agreement, the Sub-recipient assures that it will comply with applicable provisions of the following National policies on: (1) Prohibiting discrimination:

(i) On the basis of race, color, or national origin, in Title VI of the Civil Rights Act of 1964 (42 U.S.C. 2000d, et seq.), as implemented by DoD regulations at 32 CFR part 195;

(ii) On the basis of age, in the Age Discrimination Act of 1975 (42 U.S.C. 6101, et seq.) as implemented by Department of Health and Human Services regulations at 45 CFR part 90;

(iii) On the basis of handicap, in Section 504 of the Rehabilitation Act of 1973 (29 U.S.C. 794), as implemented by Department of Justice regulations at 28 CFR part 41 and DoD regulations at 32 CFR part 56;

(iv) On the basis of sex or blindness, in Title IX of the Educational Amendments of 1972 (20 U.S.C. 1681, et seq.).

(2) The Clean Air Act (42 U.S.C. 7401, et seq.) and Clean Water Act (33 U.S.C. 1251, et seq.), as implemented by Executive Order 11738 (3 CFR, 1971-1975 Comp., p. 799).

The Sub-recipient shall obtain assurances of compliance for all subawards under this effort

1.17 Prohibition on Using Funds under Grants and Cooperative Agreement with Entities that Require Certain Internal Confidentiality Agreements (January 2017)

- (a) The Sub-recipient shall not require its employees or subawardees to sign or comply with internal confidentiality agreements or statements prohibiting or otherwise restricting such employees or subawardees from lawfully reporting waste, fraud, or abuse related to the performance of a Government assistance instrument to a designated investigative or law enforcement representative of a Federal department or agency authorized to receive such information (e.g., agency Office of the Inspector General).
- (b) The Sub-Recipient shall notify current employees and subawardees that prohibitions and restrictions of any preexisting internal confidentiality agreements or statements covered by this article, to the extent that such prohibitions and restrictions are inconsistent with the prohibitions of this article, are no longer in effect.
- (c) The prohibition in paragraph (a) of this clause does not contravene requirements applicable to Standard Form 312 (Classified Information Nondisclosure Agreement), Form 4414 (Sensitive Compartmented Information Nondisclosure Agreement), or any other form issued by a Federal department or agency governing the nondisclosure of classified information.
- (d) In accordance with section 743 of Division E, Title VII, of the Consolidated and Further Continuing Appropriations Act, 2015, (Pub. L. 113-235), and its successor provisions in subsequent appropriations acts (and as extended in continuing resolutions) use of funds appropriated (or otherwise made available) is prohibited, if the Government determines that the Sub-recipient is not in compliance with the provisions of this article.

(e) The Sub-recipient shall include the substance of this clause, including this paragraph (e), in subawards.

1.18 Section 889 (August 2020)

Section 889 of the National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2019 (Public Law 115-232) prohibits the head of an executive agency from obligating or expending loan or grant funds to procure or obtain, extend, or renew a contract to procure or obtain, or enter into a contract (or extend or renew a contract) to procure or obtain the equipment, services, or systems prohibited systems as identified in section 889 of the NDAA for FY 2019.

To implement this requirement, in accordance with 2 CFR 200.216 and 200.471, DoD Grants Officers shall include a condition in all awards that are issued on or after August 13, 2020.

The condition shall include language enacting section 889 of the NDAA for FY 2019 by prohibiting the recipient and subrecipient from entering into contracts (or extending or renewing contracts) with entities that use covered telecommunications equipment or services. This prohibition shall apply even if the contract is not intended to procure or obtain any equipment, system, or service that uses covered telecommunications equipment or services.

For the purposes of this condition,

1. COVERED TELECOMMUNICATIONS EQUIPMENT OR SERVICES means any of the following:

- * For the purpose of public safety, security of government facilities, physical security surveillance of critical infrastructure, and other national security purposes, video surveillance and telecommunications equipment produced by Hytera Communications Corporation, Hangzhou Hikvision Digital Technology Company, or Dahua Technology Company (or any subsidiary or affiliate of such entities);
- * Telecommunications or video surveillance equipment or services provided by such entities or using such equipment; or
- * Telecommunications or video surveillance equipment or services produced or provided by an entity that the Secretary of Defense, in consultation with the Director of the National Intelligence or the Director of the Federal Bureau of Investigation, reasonably believes to be an entity owned or controlled by, or otherwise connected to, the government of a foreign country.

2. COVERED FOREIGN COUNTRY means the People's Republic of China.

1.19 Prohibition on a ByteDance Covered Application (June 2023)

(a) *Definitions.* As used in this article—

Covered application means the social networking service TikTok or any successor application or service developed or provided by ByteDance Limited or an entity owned by ByteDance Limited.

Information technology, as defined in 40 U.S.C. 11101(6)—

(1) Means any equipment or interconnected system or subsystem of equipment, used in the automatic acquisition, storage, analysis, evaluation, manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information by the executive agency, if the equipment is used by the executive agency directly or is used by an awardee under an

award with the executive agency that requires the use—

(i) Of that equipment; or

(ii) Of that equipment to a significant extent in the performance of a service or the furnishing of a product;

(2) Includes computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the central processing unit of a computer, software, firmware and similar procedures, services (including support services), and related resources; but

(3) Does not include any equipment acquired by a Federal awardee incidental to a Federal award.

- (b) *Prohibition.* Section 102 of Division R of the Consolidated Appropriations Act, 2023 (Pub. L. 117-328), the No TikTok on Government Devices Act, and its implementing guidance under Office of Management and Budget (OMB) Memorandum M-23-13, dated February 27, 2023, “No TikTok on Government Devices” Implementation Guidance, collectively prohibit the presence or use of a covered application on executive agency information technology, including certain equipment used by Federal awardee(s). The Awardee is prohibited from having or using a covered application on any information technology owned or managed by the Government, or on any information technology used or provided by the Awardee under this award, including equipment provided by the Awardee’s employees; however, this prohibition does not apply if the Grants Office/Agreements Officer provides written notification to the Awardee that an exception has been granted in accordance with OMB Memorandum M-23-13.
- (c) *Subcontracts/Subrecipients.* The Awardee shall insert the substance of this article, including this paragraph (c), in all subawards, including subawards for the acquisition of commercial products or commercial services.

Schedule D

Additional Requirements

Kickoff Meeting

The sub-recipient will schedule a kickoff meeting with the NCDMM Project Manager to be held within 30 days of contract award. At the kickoff meeting the sub-recipient will provide the following:

- A detailed Gantt chart for the entire project that includes clearly identified milestones. The milestones must include identification of the key elements of Technology Development, Education and Workforce, Technology Transition, Sustainability, and MRL/TRL status.
- A listing of project deliverables with associated delivery dates
- A Spend Plan by month for the entire project. The attached template is provided as assistance. The Spend Plan must incorporate both America Makes funding and cost share.
- Identification of team Sub-Recipient/Members and their roles and responsibilities

Project Reporting

For the duration of the project, monthly reports, invoices and cost share reports will be due by the 10th of the month for activities of the preceding month. The sub-recipient is responsible for assembling information from subcontractors to support reporting requirements. Both technical and financial reports will be required.

- Technical reports will highlight the reporting period's technical achievements, identify any unexpected technical issues that have been encountered and action plans to resolve such issues, state the current MRL/TRL level and justification for advancing in MRL/TRL, updated Gantt charts, progress on metrics and future planned activities. All technical reports must provide a brief publicly-releasable section and a section for release to America Makes Sub-Recipient/Members. Proprietary information available only to project team Sub-Recipient/Members and appropriate America Makes staff/Government representatives can also be contained in the report and marked accordingly. A template for the reporting format will be provided by NCDMM/America Makes.
- Financial reports and invoices will document expenditures of both America Makes funds and cost share. Pertinent backup documentation, such as cost share reports, will be required to support financial reports. Backup documentation to be supplied by the sub-recipient will include, but not be limited to, individual cost share reports for team Sub-Recipient/Members. Tracking of actual expenditures vs. those provided in the Spend Plan will be required. The ratio of America Makes funding vs. cost share funding is expected to be roughly 1:1 at all times throughout the project. Both the ratio of America Makes funding to cost share funding and actual expenditures to expenditures stated in the Spend Plan will be monitored by the NCDMM Project Manager. If either or both of these financial goals are not met, the sub-recipient will be required to submit a corrective action plan. Failure to correct financial shortcomings may be grounds for contract cancellation.
- America Makes will provide an online portal for reporting. All reports will be uploaded to the America Makes online portal site by the sub-recipient. Note: Microsoft SharePoint will be used initially and log-on credentials will be provided by the America Makes team.

Meeting Participation

- Sub-recipient participation in regularly scheduled conference calls, quarterly review meetings and Project Management Review (PMR) meetings will be required.